

Kolmogorov-Arnold Sieve Networks for High-Dimensional Estimation and Inference

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Abstract

This paper introduces Kolmogorov-Arnold Sieve Networks (KASNs), a novel deep learning framework for nonparametric estimation and semiparametric inference in high-dimensional settings. By integrating the Kolmogorov-Arnold representation theorem into a sieve extremum framework and leveraging sparsity, our proposed method achieves an explicit asymptotic rate of convergence in the root-mean-square norm of $o_P(n^{-1/4})$. The sparse compositional structure imposed on the target function generalizes multiple-index models, permitting nonlinear interactions across covariates while preserving dimension-free convergence rates. This framework delivers asymptotically normal plug-in estimators and facilitates nuisance function estimation within Double/Debiased Machine Learning (DML). These are areas where existing deep neural network architectures are less suited due to their black box nature, reliance on i.i.d. assumptions, and slower convergence rates. KASNs overcome these limitations through a learnable B-spline activation function structure that is theoretically compatible with stationary mixing processes.

Keywords: Curse of dimensionality, deep learning, double/debiased machine learning, Group Lasso, Kolmogorov-Arnold networks, method of sieves, multiple-index models, time-series.

JEL classification: C14, C32, C45, G12.

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Full code with documentation is available at: <https://github.com/samiabd8/SieveKAN>

1 Introduction

In modern statistical and economic applications, flexible nonparametric function estimation methods (such as deep neural networks, kernel-based methods, and spline-based methods, among others) are increasingly important for modeling complex relationships arising from high-dimensional datasets. A fundamental challenge facing these methods in high-dimensional settings is the curse of dimensionality (CoD), which dictates that the sample size required for a given estimation accuracy grows exponentially with the dimensionality of the covariate space. Although approaches such as generalized additive models (GAMs) seek to mitigate the CoD through structural assumptions (e.g., additivity, limited degrees of interaction), these assumptions generally fail to capture the compositional and interactive nature inherent in many complex real-world relationships. Introduced by Liu et al. (2024), Kolmogorov-Arnold Networks (KANs) represent a promising alternative to standard deep neural network (DNN) architectures that inherently addresses some of these limitations. As the name suggests, the universal approximation capabilities of KANs arise from the Kolmogorov-Arnold representation theorem (KART; Kolmogorov, 1956; Arnold 1957), which posits that continuous multivariate functions can be represented as a finite composition of univariate functions and summation operations, regardless of the degree of interaction between covariates.

This paper develops a novel and theoretically grounded sieve extremum estimator based on deep KANs for nonparametric regression in high-dimensional settings featuring time-series data. By integrating KANs with the sieve estimation framework of Chen and Shen (1998) and leveraging a sparse version of the KART (where only a small subset of univariate components are non-zero), we achieve explicit asymptotic convergence rates in the L^2 -norm that do not depend on the dimension of the covariates (d). This directly mitigates the CoD in a manner similar to sparse GAMs when the target function admits a sparse compositional form. Although we assume the true sparse compositional target function is fixed for a given sample size n , our novel regularization method centered around the Group Lasso (Yuan and

Lin, 2006) is consistent with a data-driven form of model selection from the perspective of Barron et al. (1999). The Group Lasso is a regularization technique that encourages sparsity by penalizing entire groups of coefficients simultaneously (rather than individual coordinates as in the Lasso), enabling functional selection by pruning redundant activation functions in the resulting Kolmogorov-Arnold sieve network (KASN) extremum estimator. As we discuss throughout this paper, the KASN framework can be understood as a flexible compositional extension of multiple-index models¹ (e.g., Ichimura and Lee, 1991; Hristache et al., 2001), where both the index transformations and their aggregation are nonparametric and learned from data; with sparsity governing which components are active rather than assumed a priori, as we formalize in the model assumptions. Our sieve estimation approach relies on a finite-dimensional space (Θ_n) that grows with n in order to control network complexity growth in addition to sparsity, while simultaneously ensuring sufficient network approximation power.

To illustrate, consider the nonparametric regression model $Y_t = f_0(X_t) + U_t$ where given covariates $X_t = \{X_{t,j}\}_{j=1}^d$, the parameter of interest belonging to an infinite-dimensional space Θ is defined as: $\theta_0 \equiv f_0$ with $f_0(X_t) = \mathbb{E}[Y_t|X_t]$. We approximate Θ using a sequence of finite-dimensional sieve spaces $\{\Theta_n\}$ that are dense in Θ as $n \rightarrow \infty$ (i.e., any element in Θ can be approximated arbitrarily well by an element in Θ_n for sufficiently large n). Kernel-based approaches (e.g., Nadaraya, 1964; Watson, 1964) estimate f_0 directly but suffer from the CoD as d grows; single-index restrictions of the form $f_0(X_t) = f(X'_t\beta)$ circumvent this by reducing the problem to one dimension but confine the structure to be estimated to a single linear projection (Ichimura, 1993). Fully nonparametric sieve and/or spline methods (e.g., the various estimators outlined in Chen, 2007) impose smoothness but require the researcher to pre-specify the basis structure and interaction terms. A similar tradeoff is present in multiple-index and projection pursuit methods (e.g., Friedman and Stuetzle, 1981), which extend to finitely many linear indices but still require pre-specifying the basis and index structure. GAMs of the form $f_0(X_t) = \sum_{j=1}^d f_j(X_{t,j})$ mitigate the CoD through additivity,

¹See Horowitz (2009) for a comprehensive overview of this class of models and the single-index models they generalize.

but rule out the compositional and interactive structure present in many applications. More generally, the partially linear regression (PLR) model² introduced in Robinson (1988) (which also relies on kernel-based methods) and given by $Y_t = D_t\alpha_0 + g_0(W_t) + U_t$ introduces a finite-dimensional linear component α_0 such that $\theta_0 \equiv (\alpha_0, g_0)$. Estimating α_0 at a $\sqrt{n}$ -rate requires first estimating the nuisance function g_0 at a sufficiently fast nonparametric rate, a task where the choice of method matters considerably in most settings. Unlike the aforementioned methods, the KASN extremum estimator $\hat{\theta}_n \in \Theta_n$ instead estimates the unknown function (whether f_0 in the nonparametric setting or the nuisance function g_0) as a sparse compositional function learned directly from data. Our proposed estimator delivers the $o_P(n^{-1/4})$ rate required for $\sqrt{n}$ -consistent inference on α_0 , without pre-specifying the interaction structure (see **Examples 1-2**).

In contrast to Multilayer Perceptron (MLP) architectures, a widely-used class of deep neural networks (DNNs) where approximation power is tied to network width and length (potentially leading to intractable complexity growth), the expressiveness of KASNs hinges on the flexibility of the splines characterizing the learned activation functions. MLPs represent a class of flexible nonparametric estimators that has demonstrated strong empirical performance in a variety of machine learning tasks (e.g., regression, classification, natural language processing), largely due to their powerful approximation capabilities. Relative to shallow network architectures, MLPs regularly achieve improved empirical performance (see Goodfellow et al., 2016 for a comprehensive overview). Despite this, the application of MLPs for rigorous statistical inference and econometric analysis presents significant challenges. The most critical arises from the “black box” nature of these architectures, which hinders tractability³ in a manner that models such as GAMs do not. Furthermore, much of the recent work attempting to bridge MLPs with statistical theory for inference (typically under the category of “Deep ReLU networks”, or ReLU DNNs) relies on i.i.d. assumptions,

²We formally define a partition of X_t into a treatment variable $D_t \in \mathbb{R}^{d_1}$ and controls $W_t \in \mathbb{R}^{d_2}$ in the next section.

³From the perspective of controlling the metric entropy of the network in high dimensions, which is a measure of function space complexity (e.g., in terms of covering numbers).

rendering this class of DNNs less suited for the time-series data structures common in economic/financial applications. Lastly, commonly derived convergence rates in this literature require conditions that tie the smoothness of the target function to the dimension of the covariates (imposing degrees of differentiability that are generally unrealistic for even a modest number of regressors) in order to deliver the convergence rate required for semiparametric inference, as we discuss throughout this paper.

Our proposed KASN extremum estimator $\hat{\theta}_n$ directly addresses these limitations and achieves a dimension-independent convergence rate of $o_P(n^{-1/4})$, the aforementioned rate required for delivering $\sqrt{n}$ -consistency in semiparametric inference. By verifying the necessary conditions⁴ outlined in Chen and Shen (1998), we establish convergence and the asymptotic normality of smooth (linear or otherwise) plug-in functionals of $\hat{\theta}_n$. As a result, we are able to accommodate a wide range of practical applications involving time-series and semiparametric inference. Such applications include the nonparametric regression model example that we consider in simulations, as well as treatment effect estimation in partially linear structural models. The KASN estimator consistently demonstrates strong performance across Monte Carlo simulations of these settings and an empirical application involving macroeconomic financial time-series data. In each setting, our novel regularization method centered around the Group Lasso effectively prunes redundant univariate functions in the architecture, crucial for controlling the metric entropy. We also demonstrate how our estimator can be used for nuisance function estimation in the Double/Debiased Machine Learning (DML) framework of Chernozhukov et al. (2018). DML has emerged as an indispensable tool in high-dimensional statistics for constructing valid inference for finite-dimensional parameters. We discuss how this framework offers a bias-robust alternative in ultra-high-dimensional regimes ($d \gg n$), although the development of sample-splitting for time-series data is left to future work⁵ as

⁴We demonstrate how the KASN extremum estimator satisfies these high-level assumptions (pertaining to stochastic equicontinuity and the Riesz representation theorem) from Chen and Shen (1998) by adopting the verification approach of Chen and White (1999) to KASNs.

⁵e.g., the “neighbors-left-out” cross-fitting method of Semenova et al. (2023), which would significantly complicate the independent blocks construction our main results are built upon.

it is beyond the scope of this paper.

Through the tractability of our KASN architecture featuring learned B-spline activation functions, our proposed framework bridges the gap between deep compositional networks and multiple-index models (as we demonstrate in Section 2.4). Although in theory it is possible to obtain similar dimension-free rates of convergence for existing sieve-based methods (e.g., penalized splines) by leveraging sparsity, the distinction lies in specification versus discovery. To our knowledge, no existing penalized sieve method jointly achieves: data-driven structural recovery using deep network architectures, theoretical guarantees under time-series dependence, and a convergence rate of $o_P(n^{-1/4})$. Traditional sieve methods place the burden of structural intuition on the researcher, requiring one to pre-specify the basis (including knot locations, polynomial degrees, and specific interaction terms), thus encoding structural assumptions that are often rigid in nature. In contrast, our proposed framework delegates this architectural search to the data through the learned activation functions and our aforementioned regularization method. As a result, the model recovers the active compositional components without requiring a priori knowledge of the interaction structure. This flexibility comes at the cost of significantly increased computation time relative to convex methods such as the Lasso (a tradeoff common in DNN estimation), although our implementation is designed to efficiently employ the processing power of modern computer architectures⁶. The goal of this paper is to reopen the research frontier on sieve artificial neural networks (aNNs), a foundational data-driven methodology (White, 1990; Chen and Shen, 1998; Chen and White, 1999) for flexible nonparametric estimation and semiparametric inference, as we discuss in the literature review.

1.1 Related Literature

Many of the recent contributions to the literature on DNNs revolved around networks using the Rectified Linear Units (ReLU) activation function, $\sigma(x) = \max\{0, x\}$, due to their

⁶Namely, through the use of graphics processing units (GPUs). We expand on how our proposed architecture benefits from this with respect to speeding up computation times in Appendix A.3.

convenient and relatively simple properties. The manner in which this activation function truncates negative inputs can be viewed as a natural induction of sparsity⁷ in activation maps (as opposed to in weights, see the Appendix for an illustrative example), in addition to mitigating the vanishing gradient problem (discussed later on in the context of our KASN architecture).

Yarotsky (2017, 2018) develop the approximation theory for this class of DNNs, while Farrell et al. (2021) and Schmidt-Hieber (2020) propose rigorous frameworks for their application in semiparametric inference and nonparametric regression, respectively. The former derives nonasymptotic high probability bounds for DNNs, while the latter considers a sparsely connected⁸ architecture and obtains an oracle inequality.

Linking the improved performance of networks when additional layers are added in MLPs to the KART, Schmidt-Hieber (2021) presents an interpretation of this representation theorem using deep ReLU networks under simple modifications (namely, smoothness of the target function; resulting in learnable Lipschitz continuous inner functions). This constituted a breakthrough in the application of the KART in aNNs, once thought to be irrelevant⁹ in this context. Although several papers in the time between Kolmogorov (1956) and Schmidt-Hieber (2021) have attempted to show the usefulness of the KART in aNNs (e.g., Sprecher, 1996, 1997), these were limited to network structures with two hidden layers and fixed activation functions. Unlike Schmidt-Hieber (2021), where the intuition regarding the KART is inspired by ReLU DNNs, our KASN architecture and extremum estimator are motivated by the literature concerning single-index and multiple-index models (Ichimura and Lee, 1991; Ichimura, 1993; Hristache et al., 2001). This bridges the compositional structures of DNN architectures to the econometrics literature on dimension-reduction, through shifting the

⁷Ensuring that only neurons with non-negative inputs ($x \geq 0$) remain active, preserving important features across forward passes. This differs from the *structural* sparsity relevant to this framework and methods such as weight decay.

⁸With respect to the network weights, where a compositional structure assumption is imposed on the target function.

⁹e.g., Girosi and Poggio (1989), titled: “Representation Properties of Networks: Kolmogorov’s Theorem Is Irrelevant” attributed to the otherwise fractal nature of the inner functions.

emphasis from the regularity of the inner functions in the KART to the compositions of the outer aggregations¹⁰

Applying the crucial insight from Schmidt-Hieber (2021), Liu et al. (2024) propose an alternative to MLPs in the form of KANs, which derive their universal approximation capabilities from the KART rather than the Universal Approximation Theorem (Hornik et al., 1989). While both of these DNN architectures are fully-connected, KANs replace the fixed activation functions in MLPs with learnable activation functions in the form of B-splines. The authors demonstrate how by defining the layers of KANs in terms of matrices of univariate functions, the KART can be generalized to arbitrary widths and depths. As a result, KANs are effectively models with MLPs on the outside and splines on the inside, the latter contributing to significantly improved accuracy as shown by Liu et al. (2024) in a wide range of applications. Deriving generalization bounds for KANs, Zhang and Zhou (2024) show how it is possible to guarantee that the bound scales with the L^1 -norm of the coefficients when the activation function is composed of B-spline basis functions.

In both of these KAN papers, ℓ_1 -regularization plays an essential role. While these papers lay the groundwork for KANs with learned activation functions in the form of B-splines, our proposed framework introduces a new regularization procedure centered around the Group Lasso ($\ell_{1,2}$ -regularization, targeting groups of coefficients *across* activations) and achieves dimension-free rates of convergence for the KASN M-estimator. Gouriéroux and Monfort (2025) formalize the identification issues and representation non-uniqueness inherent in Kolmogorov-Arnold architectures¹¹ and relate multi-layer networks to stochastic state-space models.

In a major contribution to the literature on sieve extremum estimation (Grenander, 1981), Chen and Shen (1998) obtain convergence rates for these nonparametric and semiparametric estimators involving time-series data; in addition to establishing the $\sqrt{n}$ -consistency and asymptotic normality of plug-in sieve extremum estimates with respect to smooth function-

¹⁰As we demonstrate in the discussion of Assumptions **K.3(a)** and **K.4(c)**.

¹¹Directly addressed by the KASN framework throughout this paper.

als¹². The method of sieves entails estimating a function belonging to a possibly infinite-dimensional parameter space Θ using a sequence of approximating parameter spaces (growing in complexity as $n \rightarrow \infty$, defined as the sieve space Θ_n) that are dense in Θ . This sieve M-estimator framework derives theoretical results using a quasi-maximum-likelihood estimation (QMLE) method (White, 1982; Gouriéroux et al., 1984) which does not suffer from the issues associated with infinite-dimensional maximum likelihood (namely, slow convergence rates and potential inconsistency under misspecification). Crucially, the extension of sieve extremum estimators to time-series data in Chen and Shen (1998) allows for the application of this method to a wide range of economic applications that would otherwise not be possible under the i.i.d. assumptions made in the existing literature (e.g., Shen, 1997). Among the applications involving time-series models considered to demonstrate this sieve extremum estimator, rates of convergence for nonlinear state-space models using aNNs are obtained. Chen and White (1999) obtain improved approximation rates (explicitly $o_P(n^{-1/4})$) for shallow sieve aNN estimators building directly upon this framework, see the Appendix for a full exposition. An outline of regularization approaches in nonlinear (finite-dimensional or otherwise) sieve estimation can be found in Chen (2007). Shen et al. (2023) investigate the asymptotic properties of shallow sieve network estimators and discuss how their results can potentially be extended to DNNs with Lipschitz continuous activation functions. To the best of our knowledge, there does not exist a framework integrating the method of sieves with modern DNN architectures in order to establish the consistency of such networks in nonparametric regression.

The remainder of this paper is organized as follows. Section 2 defines the target function space and outlines our proposed KASN extremum estimator, including the key parameters of the KASN architecture and its assumptions. Section 3 presents our main theoretical results on approximation error and convergence rates. We present results of our Monte Carlo

¹²A “plug-in” (or substitution) estimator is a method of estimating a functional of interest by substituting unknown population distributions or nuisance functions with their estimated counterparts; see Shen (1997) for a general overview.

simulations for our DML (semiparametric mean regression) model in addition to a simple nonparametric regression in Section 4. Section 5 features our two empirical applications involving macroeconomic financial variables, including a forecasting exercise. We conclude with a summary and a discussion of directions for future research in Section 6.

2 Kolmogorov-Arnold Sieve Networks

We begin by defining the relevant metric spaces and their properties, in addition to the notation used throughout this paper. Although we adopt the notation of Chen and White (1999), our approach diverges in the characterization of the target function space. Rather than assuming functions in this space admit a Fourier representation¹³, we impose the more standard assumption that it takes on the form of a Sobolev space. As we discuss in Section 2.4, we impose a comparable yet local structural assumption on the univariate functions (Lipschitz continuity), rather than the global spectral assumption of Chen and White (1999).

2.1 Notation

Given a strictly stationary sequence $\{Z_t\}_{t=1}^n = (Y_t, X_t')'$ with dimension $(1 + d)$ and joint probability measure P_0 , where $\{X_t\}_{t=1}^n$ is a sequence of covariates (potentially including lags), let μ be a probability measure on $\mathbb{R}^d$ (the marginal distribution of $X_t \in \mathbb{R}^d$). For the semiparametric structural models of interest, define the partition $X_{t,\mathcal{D}} \in \mathbb{R}^{d_1}$ and $X_{t,\mathcal{X}} \in \mathbb{R}^{d_2}$, where $\mathcal{D} = \{1, \dots, d_1\}$ and $\mathcal{X} = \{d_1 + 1, \dots, d\}$ and $d_1 + d_2 = d \in \mathbb{N}$. For simplicity, let $D_t \equiv X_{t,\mathcal{D}}$ (representing the linear component) and $W_t \equiv X_{t,\mathcal{X}}$ (noting that μ now corresponds to $W_t \in \mathbb{R}^{d_2}$) in semiparametric applications.

We define $L_2(\mu)$ as the Hilbert space of all measurable functions $h : \mathbb{R}^d \rightarrow \mathbb{R}$ such that

¹³Central to obtaining dimension-independent rates of convergence in their framework, see Section A.5 in the Appendix for a full exposition.

$\int_{\mathbb{R}^d} |h(x)|^2 d\mu(x) < \infty$, equipped with the following norm:

$$\|h\|_{L_2(\mu)} = \left(\int_{\mathbb{R}^d} |h(x)|^2 d\mu(x) \right)^{1/2}$$

To simplify the notation throughout this paper, let $\|\cdot\| \equiv \|\cdot\|_{L_2(\mu)}$ unless stated otherwise. In order to characterize the space of the target function ($f_0 \equiv \theta_0 \in \Theta$ for non-parametric regression¹⁴, where Θ may more generally represent a class of regression functions, probability densities, and other structural mappings of interest), we define the multi-index¹⁵ $\bar{\lambda} = (\bar{\lambda}_1, \dots, \bar{\lambda}_d)^\top$, which represents a vector of non-negative integers with order $|\bar{\lambda}| = \sum_{i=1}^d \bar{\lambda}_i$. Let $D^{\bar{\lambda}}$ represent the weak partial derivative operator corresponding to $\bar{\lambda}$, where:

$$D^{\bar{\lambda}}h(x) = \frac{\partial^{|\bar{\lambda}|}h(x)}{\partial x_1^{\bar{\lambda}_1} \dots \partial x_d^{\bar{\lambda}_d}}$$

The target function space is defined as a Sobolev space $\mathcal{F}_d^{m+1} \equiv W_2^{m+1}(\mu)$. In addition to consisting of all measurable functions $h : \mathbb{R}^d \rightarrow \mathbb{R}$ such that $h \in L_2(\mu)$, this space contains its weak derivatives $D^{\bar{\lambda}}h \in L_2(\mu)$ for all multi-indices $\bar{\lambda}$ with $|\bar{\lambda}| = \sum_{i=1}^d \bar{\lambda}_i \leq m+1$. The norm for a Sobolev space $W_2^{m+1}(\mu)$ is defined as:

$$\|h\|_{W_2^{m+1}(\mu)} = \left[\sum_{|\bar{\lambda}| \leq m+1} (\|D^{\bar{\lambda}}h\|_{L_2(\mu)})^2 \right]^{1/2}$$

where $m \geq 0$ is an integer governing the smoothness of the target function $f_0 \in \mathcal{F}_d^{m+1}$. Specifically, the value $m+1$ is the upper bound on the order of the multi-index¹⁶ $\bar{\lambda}$ dictating the maximum order of partial derivatives that are required to be square-integrable (i.e., belonging to $L_2(\mu)$).

¹⁴Whereas $\theta_0 \equiv (\alpha_0, f_0)$ in the aforementioned semiparametric applications, e.g., the partially linear regression model (Robinson, 1988).

¹⁵A vector specifying the number of partial derivatives to be taken with respect to each corresponding input variable.

¹⁶For example, when $d = 2$ and $m = 0$, the norm sums derivatives of order 0 (the function $h \in L_2(\mu)$ itself, where $\bar{\lambda} = (0, 0)$) and order 1 (where $\bar{\lambda} = (1, 0)$ or $\bar{\lambda} = (0, 1)$, representing $\frac{\partial h}{\partial x_1}$ and $\frac{\partial h}{\partial x_2}$). This yields three terms in total.

As we demonstrate in our model assumptions, this definition allows us to quantify the smoothness of the true function we seek to estimate. As a result, we are able to link the properties of the proposed KASN architecture that features (learned) spline activation functions to this function space. Critically, this target space allows us to work towards convergence rates where the dependence on the smoothness parameter m is absorbed into constants; unlike typical “minimax” optimal convergence rates¹⁷ (Stone, 1982) commonly found in the literature on deep ReLU networks. For example, a common minimax rate of convergence in nonparametric estimation takes on the form of $n^{-m/(2m+d)}$, which is prohibitively slow in many potential high-dimensional settings. By avoiding such a dependency, our approach is a key step towards mitigating the CoD in DNN architectures applied in semiparametric inference.

Although our analysis relies on the smoothness provided by the Sobolev space, we are only able to obtain convergence rates in the L^2 -norm rather than in the stronger Sobolev norm (the latter implying convergence in the former). We will demonstrate why this is the case when applying spline approximation theory in our key results, as rates of convergence in the Sobolev norm would not be independent of m under the KASN architecture. Nonetheless, this weaker form of convergence is sufficient for the applications we consider, as well as for important extensions¹⁸. Furthermore, given how quickly convergence rates involving non-static conditions on m breakdown in high-dimensional settings, this tradeoff is justified by how it enables the proper theoretical grounding for analysis in these settings.

We assume the target function $f_0 \in \mathcal{F}_d^{m+1} \equiv W_2^{m+1}(\mu)$ admits a sparse compositional structure (Assumption **K.3(a)**), where only a small subset (of cardinality s_0) of the univariate functions in the KART are non-zero. This form of sparsity arises naturally in many settings involving high-dimensional data and we thoroughly verify it computationally across Monte Carlo simulations and our empirical applications. For example, in a financial risk

¹⁷Where the smoothness degree of the target function m is tied to d ; see Schmidt-Hieber (2020) for a full discussion in relation to DNNs.

¹⁸Namely, obtaining asymptotic normal linear plug-in estimators, where the nonparametric component is estimated using our proposed KASN extremum estimator.

model predicting the probability of a given borrower defaulting, a researcher might collect hundreds of variables based on available data. It is reasonable to expect that only a few covariates (e.g., including payment history and debt-to-income ratios) truly impact odds of default, with complex and not necessarily additive nonlinear relationships arising between this small subset of variables. Formally, the KART states:

Definition 1 (Kolmogorov-Arnold Representation Theorem): for a multivariate continuous $f : [0, 1]^d \rightarrow \mathbb{R}$ and given univariate functions $\phi_{i,j} : \mathbb{R} \rightarrow \mathbb{R}$, $\Phi_i : \mathbb{R} \rightarrow \mathbb{R}$, we have:

$$f(X_t) = f(X_{t,1}, X_{t,2}, \dots, X_{t,d}) = \sum_{i=1}^{2d+1} \Phi_i \left(\sum_{j=1}^d \phi_{i,j}(X_{t,j}) \right) \quad (2.1)$$

where the inner functions $\phi_{i,j}$ represent univariate feature transformations of each input $X_{t,j}$; the inner sum $\sum_{j=1}^d \phi_{i,j}(X_{t,j})$ aggregates these into a scalar summary (which can be viewed a nonlinear generalization of the linear index $X_t' \beta$ in single-index models) and Φ_i acts as a nonlinear link function on that summary. Together, this two-level structure reduces a d -dimensional estimation problem to at most $2d + 1$ one-dimensional problems, regardless of the interaction structure of $f(\cdot)$. As we discuss below, our sparse variant (Assumption **K.3(a)**) uses only $s_0 \ll 2d + 1$ active components, which is central to mitigating the CoD. In order to ensure regularity of these univariate components, we impose smoothness and Lipschitz continuity assumptions (**K.4(a)-(b)**) in addition to the smoothness of the target function $f_0 \in \mathcal{F}_d^{m+1}$. In contrast to the UAT...

In the context of the possibly infinite-dimensional parameter space $\mathcal{F}_d^{m+1}$, this sparse compositional assumption is analogous to those made in sparse GAM frameworks¹⁹, as it entails approximating a multivariate function using a subset of univariate functions²⁰. As we demonstrate throughout this paper, a KASN can be viewed as a generalization of these models to sparse compositional structures (thus allowing for complex interactions), embed-

¹⁹Such as Raskutti et al. (2011), Meier et al. (2009), and Friedman (1991).

²⁰Also referred to one dimensional, or “1D” functions.

ded within a DNN architecture. In terms of multiplicative interactions, Liu et al. (2024) demonstrate how even a simple two-layer KAN structure is capable of approximating functions such as $f(x_1, x_2) = x_1 x_2$. In this example, it can be observed that the KAN computes $x_1 x_2$ by leveraging $2x_1 x_2 = (x_1 + x_2)^2 - (x_1^2 + x_2^2)$, where the spline functions learn the individual parts of this decomposition. The main consequence of this sparse compositional assumption imposed on the target function $f_0 \in \mathcal{F}_d^{m+1}$ is that it allows for the dimension of the covariates, d , to grow with the sample size at even a polynomial rate: $d = O(n^{\zeta_d})$ for any $\zeta_d > 0$.

The main consequence of this structural assumption is that it facilitates achieving explicit asymptotic convergence rates in the L^2 -norm that do not depend on d at all. The ability of our proposed framework to permit covariate/input dimensions that scale as the sample increases without causing the metric entropy to grow uncontrollably²¹ makes it a very promising direction for the application of sparse deep networks in high-dimensional settings; namely due to how assumptions commonly made in high-dimensional models such as $s_0 = O(1)$ (which implies that the target function can be approximated by a fixed number of univariate components: e.g., $f(X_t) = \sum_{j=1}^{s_0} f_j(X_{t,j})$ in GAMs) can be overly restrictive²² in many practical applications. In GAMs, s_0 represents the number of covariates that have a non-zero functional effect on the response variable. As a result, we mitigate the CoD by exploiting sparsity in a manner similar to sparse additive models and bound the network entropy by using a sieve estimation approach. We note that a similar claim is also made in Liu et al. (2024), one that is debatable for general KAN classes in the absence of sparsity and appropriate mechanisms for controlling complexity²³. Without these theoretical underpinnings, the network complexity *implicitly scales with* d and renders general KAN classes vulnerable to exponential growth of said complexity, especially in high-dimensional

²¹By leveraging spline approximation theory, in addition to the method of sieves and structural sparsity, as we discuss in the next section.

²²To the contrary, we allow s_0 to grow slowly in tandem with the *size* of the KASN architecture as $n \rightarrow \infty$.

²³e.g., the sparse compositional form of the target function central to our framework (Assumption **K.3(a)**).

settings. Going forward, we use the symbol $\asymp$ to denote asymptotic equivalence²⁴, which is used throughout the network parameter definitions. Furthermore, let $\mathbb{E}[\cdot]$ and $P(\cdot)$ denote the expectation and probability under measure P_0 (the same applies to $Var[\cdot]$ and $Cov[\cdot]$) henceforth.

2.2 KASN Estimator

We integrate our proposed KASN architecture in the sieve extremum estimator framework of Chen and Shen (1998), which allows for time-series data (Assumption **K.1**) and avoids the reliance on the otherwise restrictive i.i.d. assumption (e.g., as in Schmidt-Hieber, (2020) and Farrell et al., (2021), among many other DNN frameworks designed for nonparametric regression). This is of significance as a wide range of the potential high-dimensional applications a researcher might be interested in using DNNs for generally involve time-series data. In addition to obtaining asymptotic normal plug-in estimators, we consider semiparametric applications in the double/debiased machine learning framework introduced by Chernozhukov et al. (2018). Our proposed KASN extremum estimator²⁵ achieves the required rate of convergence (precisely $o(n^{-1/4})$ in the L^2 -norm) for nuisance function estimation under Neyman-orthogonal moments/scores in DML.

For the strictly stationary process $\{Z_t\}_{t=1}^n = (Y_t, X_t)'$ and the true probability measure P_0 , where $\theta_0 : \mathbb{R}^d \rightarrow \mathbb{R}$ is defined as the maximizer of the population expected criterion function (identified by the functional $\mathcal{T}$ under P_0), we define the following:

$$\theta_0 \equiv \mathcal{T}(P_0) \equiv \arg \sup_{\theta \in \Theta} \mathbb{E}[l(Z_t, \theta)] \quad (2.2)$$

$$\hat{\theta}_n \equiv \arg \sup_{\theta \in \Theta_n} \frac{1}{n} \sum_{t=1}^n l(Z_t, \theta) \equiv \arg \sup_{\theta \in \Theta_n} \mathcal{L}_n(\theta) \quad (2.3)$$

²⁴Where $a_n \asymp b_n \iff a_n \gtrsim b_n$ and $b_n \gtrsim a_n$ as $n \rightarrow \infty$. This is equivalent to $b_n = O(a_n)$ and $a_n = O(b_n)$.

²⁵Following Chen and White (1999), we refer to θ_n as such due to how the sieve space Θ_n is constructed using the outputs of our deep sieve network architecture.

Here, $l(\cdot, \cdot)$ is a quasi-log-likelihood function²⁶ belonging to a class of general loss functions, permitting a variety of nonparametric/semiparametric regression problems. For example, in nonparametric mean regression ($f_0 \equiv \theta_0$) we have $l(Z_t, \theta) = -\frac{1}{2}(Y_t - \theta(X_t))^2$, where the solution to the population problem is the conditional expectation $\theta_0 = \mathbb{E}[Y_t | X_t = x]$ given $\mathbb{E}[\epsilon_t | X_t] = 0$. In the case of nonparametric quantile regression (where the indicator-weighted loss is equivalent to the check loss; see Koenker, 2005), $l(Z_t, \theta) = -|Y_t - \theta(X_t)|[\tau \mathbf{1}_{\{Y_t \geq \theta(X_t)\}} + (1 - \tau) \mathbf{1}_{\{Y_t < \theta(X_t)\}}]$, where for the τ -th quantile and given the inverse CDF $F^{-1}(\cdot)$, we have the maximizer $\theta_0(x) = F_{Y|X}^{-1}(\tau | x)$ satisfying $P(Y_t \leq \theta_0(x) | X_t = x) = \tau$.

The following examples preview the main results of this framework. Formal assumptions are stated in Section 2.4 and the reader is referred there for precise conditions (Eq. 2.7-2.9). For nonparametric regression models, $\hat{\theta}_n \in \mathcal{K}_n$ denotes the maximizer in the KASN class defined formally in Section 2.3 (see Eq. 2.5). The $o_P(n^{-1/4})$ rate of convergence that is central to this framework is established in **Corollary 3**, which can be found in Section 3.

Example 1 (Partially Linear Regression Model). *Suppose $\{Y_t, D_t, W_t'\}$ satisfies the model²⁷*

$$Y_t = D_t \alpha_0 + g_0(W_t) + U_t, \quad \mathbb{E}[U_t | D_t, W_t] = 0,$$

where D_t is the (treatment) variable of interest and g_0 is an unknown nonparametric function of controls W_t . Assume $g_0 \in W_2^{m+1}$ admits a sparse compositional form (Assumption **K.3(a)**), where $\theta \equiv (\alpha_0, g_0) \in \Theta$ and μ is the distribution of W_t . Let $\hat{g}_n \in \mathcal{K}_n$ be the KASN estimator for g_0 , and let $\hat{\alpha}_n$ be the resulting plug-in estimator of α_0 obtained by partialling out $\hat{g}_n$ (Robinson, 1988). Then, under Assumptions **K.1-4** for $\hat{\theta}_n \equiv (\hat{\alpha}_n, \hat{g}_n)$, we have:

- (i) $\|\hat{g}_n - g_0\|_{L_2(\mu)} = o_P(n^{-1/4})$, and
- (ii) $\sqrt{n}(\hat{\alpha}_n - \alpha_0) \rightsquigarrow \mathcal{N}(0, \Sigma_\alpha)$,

²⁶A real-valued mapping $l : \mathcal{Z} \times \Theta \rightarrow \mathbb{R}$ that scores how well a given θ fits an observation Z_t , without requiring correct specification of the full likelihood.

²⁷See the partition of X_t defined earlier in the context of semiparametric structural models.

where Σ_α is the asymptotic variance characterized in **Theorem 4**.

Example 2 (Average Partial Derivative Estimation). Suppose $\{Y_t, X_t'\}$ satisfies

$$Y_t = f_0(X_t) + U_t, \quad \mathbb{E}[U_t|X_t] = 0, \text{Var}[U_t|X_t] = \sigma^2(X_t) > 0$$

with $\theta_0 \equiv f_0 \in W_2^{m+1}$ admitting a sparse compositional form²⁸. Let $\hat{\theta}_n \in \mathcal{K}_n$ be the KASN extremum estimator obtained using $l(Z_t, \theta) = -(Y_t - \theta(X_t))^2/2$. For a covariate $X_{t,j}$ with $j \in \{1, \dots, d\}$ and under Assumptions **K.1-4** with $k = 4$:

$$\begin{aligned} (i) \quad & \|\hat{\theta}_n - \theta_0\|_{L_2(\mu)} = o_P(n^{-1/4}), \text{ and} \\ (ii) \quad & \sqrt{n} \left(\frac{1}{n} \sum_{t=1}^n \frac{\partial \hat{\theta}_n(X_t)}{\partial x_j} - \mathbb{E} \left[\frac{\partial \theta_0(X_t)}{\partial x_j} \right] \right) \rightsquigarrow \mathcal{N}(0, \Sigma_{\gamma_j}). \end{aligned}$$

where Σ_{γ_j} is long-run variance of the score process $\partial \theta_0(X_t)/\partial x_j$.

Example 3 (Dynamic Discrete Choice Model Estimation). Consider a dynamic discrete choice model²⁹ where an agent chooses an action $d_t \in \{1, \dots, J\}$ each period to maximize expected discounted utility. The per-period utility is $u(s_t, d_t; q)$, where s_t is the state vector and q is a vector of parameters. The value function satisfies:

$$V(s_t; q) = \max_{d_t} \{u(s_t, d_t; q) + \beta \mathbb{E}[V(s_{t+1}; q) \mid s_t, d_t]\},$$

with discount factor $\beta \in (0, 1)$. Let $F(s_t, d_t; q) \equiv \mathbb{E}[V(s_{t+1}; q) \mid s_t, d_t]$ be the expected value function. Assume further that F_0 admits a sparse compositional representation as in Assumption **K.3(a)** (e.g., the utility function $u(s_t, d_t; q)$ is additively separable across states).

²⁸Where $\mathbb{E}[\partial \theta_0(X_t)/\partial x_j]$ generalizes the average derivative estimand of Powell et al. (1989) beyond the single-index model setting to the sparse compositional structure in Assumption **K.3(a)**.

²⁹See Norets (2012) for an example of the application of sieve aNNs in this setting. An extension of Norets (2012) centered around inference procedures facilitated by the KASN extremum estimator is left to future work. In this example, the KASN estimator permits time-series dependence under Assumption **K.1**, unlike the aforementioned application.

Let $\hat{F}_n \in \mathcal{K}_n$ be the KASN estimator obtained using $l(Z_t, F) = -(V(s_{t+1}; q) - F(s_t, d_t; q))^2/2$ (where $V(s_{t+1}; q)$ is obtained as in Norets, 2012). Then under Assumption **K.1-4**, we have:

$$\begin{aligned} (i) \quad & \|\hat{F}_n - F_0\|_{L_2(\mu)} = o_P(n^{-1/4}), \text{ and} \\ (ii) \quad & \sqrt{n} \left(\frac{1}{n} \sum_{t=1}^n \hat{F}_n(s_t, d_t; q) - \mathbb{E}[F_0(s_t, d_t; q)] \right) \rightsquigarrow \mathcal{N}(0, \Sigma_F), \end{aligned}$$

where μ is the joint distribution of (s, d) and Σ_F is the asymptotic variance.

2.3 KASN Architecture

The network architecture of our proposed KASN has three main parameters: the number of layers (L_n), the network width (W_n ; corresponding to the number of nodes W_n^l in a given layer $l \in [1, \dots, L_n]$) and the size of the B-spline grid (G_n) for the learnable activation functions which makes the network fully tractable. Unlike how MLPs place fixed activation functions on nodes, the learnable activation functions belong on *edges* (or, “in-between” layers) so that the KAN nodes simply perform summation. We define $r_n \equiv L_n W_n^2$ as a key parameter³⁰ representing the number of total *potential* edges in our KASN architecture. Central to controlling the metric entropy growth (see the bound established in **Theorem 2**) in this framework is obtaining a subset of *active* edges (s_n) through our regularization procedure.

Due to how the expressiveness KANs are instead linked to the flexibility of the splines (i.e., is capable of being increased by simply expanding the grid size), this architecture facilitates the controlled growth of the sieve space. In these structures, the width determines the number of interactions between input variables and by extension, the diversity of interactions within a given layer. This makes KANs well-suited for sieve estimation in high-dimensions, in addition to how they do not suffer from the black box nature of MLPs. Likewise, stacking

³⁰Where in the case of the shallow sieve networks from Chen and White (1999), $r_n \equiv W_n$ simply represents the number of hidden units, as outlined in Section A.5 of the Appendix.

additional layers³¹ in a KAN permits richer representations of f_0 . To this end, Schmidt-Hieber (2020) points out that allowing the length of deep networks to grow concurrently with the sample size is intuitive. Similarly, the concluding remarks of Hornik et al. (1989) addresses the natural connection between multilayer feedforward networks and the method of sieves. The tractability of KAN architectures significantly contributes to this framework that harmonizes DNNs with a method of sieves estimation approach for nonparametric regression and semiparametric inference.

To ensure the domain is bounded³² between $[0, 1]$, a CDF normalization is applied to inputs $X_t \equiv [X_{t,1}, \dots, X_{t,d}]' \in \mathbb{R}^d$ (inducing compact support on the probability measure, a common assumption in sieve estimation frameworks) such that:

$$\tilde{X}_{t,j} = \hat{F}_n^j(X_{t,j}), \quad j = 1, \dots, d \quad (2.4)$$

where $\hat{F}_n^j(\cdot)$ is the (marginal) empirical distribution function (EDF). By Theorem 19.3 of van der Vaart (1998), this introduces an approximation error of $O(n^{-1/2})$: a term negligible with respect to the resulting KASN approximation error established in **Theorem 1**. Although commonly applied in empirical finance applications (e.g., copula models), the use of CDF normalization of inputs in deep learning architectures remains underexplored. Employing this normalization in the place of commonly used standardization techniques facilitates the use of a fixed spline grid; define $\mathcal{I}_\epsilon \equiv [-\epsilon, 1 + \epsilon]$ for a small $\epsilon > 0$ padding term, so that $[0, 1] \subset \mathcal{I}_\epsilon$ by construction. WLOG, let $\mu \equiv \tilde{\mu}$ denote the distribution of the normalized inputs $\tilde{X}_t \in [0, 1]^d$, with all univariate function spaces are defined on $\mathcal{I}_\epsilon$ henceforth.

Let $\Theta \equiv A \times \mathcal{F}_d^{m+1}$ be the parameter space where $A \subset \mathbb{R}^a$ is a compact set representing the finite-dimensional space of our linear parameters of interest and Θ is equipped with the pseudometric $\|\cdot\|$. Given B-spline order $k \geq m + 1$ and coefficient vector $\mathbf{c}_n \equiv \{\mathbf{c}_j\}_{j=1}^{r_n}$, the

³¹Where each layer is a matrix of univariate functions Φ_l ; see Appendix A.3 for an extended comparison with MLP architectures.

³²e.g., as required by the KART.

finite-dimensional sieve space is given by $\Theta_n \equiv A \times \mathcal{K}_n$, where the KASN class is defined as:

$$\mathcal{K}_n = \left\{ f(\tilde{X}_t; \mathbf{c}_n) \mid f(\mathbf{x}; \mathbf{c}_n) = \Phi_{L_n} \circ (\Phi_{L_n-1} + h) \circ \dots \circ (\Phi_1 + h) \circ \Phi_0(\mathbf{x}), \right. \\ \left. \mathbf{c}_n \in \mathbb{R}^{r_n \times G_n}, \quad \sum_{j=1}^{r_n} \|\mathbf{c}_j\|_2 \leq \Delta_n \right\}, \quad (2.5)$$

where $h : \mathbb{R}^{W_n^{l-1}} \rightarrow \mathbb{R}^{W_n^l}$ is the identity residual connection³³ $h(x) = x$, each $\Phi_l : \mathbb{R}^{W_n^{l-1}} \rightarrow \mathbb{R}^{W_n^l}$ (with $\Phi_0 : [0, 1]^d \rightarrow \mathbb{R}^{W_n^1}$ and $\Phi_{L_n} : \mathbb{R}^{W_n^{L_n}} \rightarrow \mathbb{R}$) is a layer of univariate B-spline activation functions with coefficients $\mathbf{c}_j = (c_{j,1}, \dots, c_{j,G_n})' \in \mathbb{R}^{G_n}$ for each potential edge $j \in \{1, \dots, r_n\}$, and Δ_n is the B-spline coefficient constraint defined in (2.8). By the Heine-Borel theorem, the compactness of $\mathcal{K}_n$ (and by extension, Θ_n) follows as the continuous image of a closed and bounded subset of $\mathbb{R}^{r_n \times G_n}$ defined by the $\ell_{1,2}$ -constraint $\sum_{j=1}^{r_n} \|\mathbf{c}_j\|_2 \leq \Delta_n$. Explicitly, each univariate edge takes on the form of $\phi(x) = \sum_{g=1}^{G_n} c_g B_g(x)$.

We define the following adaptive threshold for what constitutes an active edge: Δ_n/r_n , which corresponds to a baseline network where coefficients are evenly distributed³⁴ across all potential edges. This tractable and data-adaptive criterion addresses the potential issues associated with arbitrary fixed thresholds commonly applied in KAN architectures, particularly in the presence of many covariates. We note that it is possible for activation functions with non-zero coefficients that do not belong to the active paths (i.e., edges disconnected from the network output) to be pruned. See Figure 1 in the Appendix (A.3) for an example of such a function, corresponding to the first input edge. In order to highlight the robustness of our regularization approach centered around the constrained set of coefficients $\sum_{j=1}^{r_n} \|\mathbf{c}_j\|_2 \equiv \|\mathbf{c}_n\|_{1,2}$, we only prune edges falling below the Δ_n/r_n threshold. The vast ma-

³³A parameter-free mapping that supports gradient stability by bounding gradients away from zero, in a manner similar to ReLU activations in MLPs. Note that the KASN width W_n is fixed across every hidden layer $\ell \in [2, \dots, L_n]$. See Section A.3 in the Appendix for an expanded discussion, including a formal definition of the recursive B-spline functions.

³⁴See Appendix A.5 for an exposition on entropy regularization, a method frequently applied in reinforcement learning (Ahmed et al., 2019) in order to encourage exploration via selection of increasingly stochastic policies. We discuss how this method is used heuristically in KAN implementations (Liu et al., 2024) and contrast it with our theoretically grounded regularization method.

jority of active edges after training contribute directly to the network output across Monte Carlo simulations and empirical applications, justifying this conservative choice. Throughout our theoretical results, we verify that this pruning method based on this adaptive threshold is compatible with the asymptotic rates of convergence achieved by the KASN estimator, provided s_n satisfies Assumption **K.3(b)**.

The KASN estimator is defined as the constrained maximizer $\hat{\theta}_n = \arg \sup_{\theta \in \mathcal{K}_n} \mathcal{L}_n(\theta)$, where $\mathcal{K}_n$ (which given Δ_n , is a deterministic class of functions) imposes the $\ell_{1,2}$ -constraint $\|\mathbf{c}_n\|_{1,2} \leq \Delta_n$ with radius $\Delta_n \asymp n^{\zeta_\delta}$ (see (2.8)). In practice, optimization is performed using the equivalent penalized objective function $\mathcal{L}_n(\theta) - \lambda \|\mathbf{c}_n\|_{1,2}$, where $\hat{\mathbf{c}}(\lambda)$ is used to denote the maximizer for a given penalty weight $\lambda > 0$ (not to be confused with our multi-index definition earlier). By Lagrangian duality³⁵, the constrained and penalized optimization problems are equivalent at the solution $\|\hat{\mathbf{c}}(\lambda_n)\|_{1,2} = \Delta_n$. For a given Δ_n , the corresponding penalty weight is therefore:

$$\lambda_n \equiv \inf\{\lambda > 0 : \|\hat{\mathbf{c}}(\lambda)\|_{1,2} \leq \Delta_n\} \quad (2.6)$$

where λ_n can be selected via cross-validation (CV) over the monotone (non-increasing) correspondence $\lambda \mapsto \|\hat{\mathbf{c}}(\lambda)\|_{1,2}$, which is equivalent to selecting Δ_n (through the hyperparameter $\zeta_\delta > 0$) using CV. The asymptotic theory in Section 3 is developed for any deterministic sequence $\Delta_n \asymp n^{\zeta_\delta}$ with a fixed $\zeta_\delta > 0$; the use of CV in finite-samples does not affect the asymptotic properties of $\hat{\theta}_n$ through the aforementioned equivalence of the optimization problems. In practice, we apply holdout validation (see Tashman, 2000 for an overview) and leave the development of a compatible *hv*-block CV procedure³⁶ based on Racine (2000) to

³⁵Under the regularity and identification conditions in Assumptions **K.2(b)-(c)**. See Section 2.2 of Shen (1997) for a discussion on penalization in the context of *approximate* sieve estimators with non-convex spaces. This includes the condition $\mathcal{L}_n(\hat{\theta}_n) \geq \sup_{\theta \in \Theta_n} \mathcal{L}_n(\theta) - O(\tilde{\epsilon}_n^2)$ with $\tilde{\epsilon}_n \rightarrow 0$ as $n \rightarrow \infty$, which is satisfied by any measurable maximizer $\hat{\theta}_n$ here: the compactness of $\mathcal{K}_n$ and continuity of $\mathcal{L}_n(\theta)$ via Assumption **K.2(c)** jointly guarantee exact attainment of the supremum by the extreme value theorem.

³⁶A method that entails eliminating h observations as a buffer to eliminate serial correlation and uses a block of v observations as a test set.

future work (in addition to the aforementioned sample splitting extension to the stationary mixing sequences in this framework, in order to keep a focused scope). We demonstrate the robustness of holdout validation (standard in time-series forecasting applications) through a sensitivity analysis reported in the Appendix.

2.4 Key Parameters and Assumptions

For cases where A is a nonempty set (e.g., in Example 1), we have: $\theta_0 \equiv (\alpha_0, f_0) \in \Theta$ and $\theta_n \equiv (\alpha_n, f(\cdot; \mathbf{c}_n)) = (\alpha_n, \pi_n f_0) \in \Theta_n$, where π_n is a projection operator representing the best possible approximation of f_0 in the sieve space. Specifically, π_n is the orthogonal projection under the L^2 -norm which ensures that the approximation is the unique element in the sieve space closest to the true θ_0 : $\pi_n \theta_0 \equiv \arg \min_{\theta \in \Theta_n} \|\theta - \theta_0\|$. Existence follows from the compactness of $\mathcal{K}_n$ and the continuity of the L^2 -norm; uniqueness follows from the strict convexity of this norm. Under the pseudometric³⁷ $\|\cdot\|$ on Θ , we define Θ_n as dense in Θ : for any $\theta \in \Theta$, there exists a $\pi_n \theta \in \Theta_n$ such that $\|\theta - \pi_n \theta\| \rightarrow 0$ as $n \rightarrow \infty$. This ensures that the approximation error (**Theorem 1**) induced by the sieve space Θ_n becomes asymptotically negligible relative to the estimation error (**Theorem 3**).

We define the KASN architecture parameters³⁸ $p_n \equiv (L_n, W_n, G_n)$ as follows:

$$L_n \asymp \log n, \quad W_n \asymp s_0 \asymp \log n, \quad G_n \asymp n^\gamma \quad (2.7)$$

$$\Delta_n \asymp n^{\zeta_\delta}, \quad \zeta_\delta > 0, \quad \frac{1}{4k} < \gamma < \frac{1}{2} \quad (2.8)$$

where γ is a parameter required to guarantee sufficient approximation power (using B-spline activation functions of order $k > 0$) in order to deliver our $o_P(n^{-1/4})$ convergence rate. In order to reflect the deep compositional structure of the KASN architecture (Assumptions

³⁷Recall that a pseudometric satisfies the axioms of a metric with the exception that $\|\theta_1 - \theta_2\|$ does not necessarily imply that $\theta_1 = \theta_2$. In the method of sieves literature, $\|\cdot\|$ is typically used interchangeably with $d(\cdot, \cdot)$.

³⁸More specifically, under B-spline order k : $p_n = \sum_{l=0}^{L_n-1} W_l W_{l+1} (G_n + k - 1) \approx L_n W_n^2 G_n = r_n G_n$.

K.3-4), we define the true total active edge count $\bar{s}_0$ as follows:

$$\bar{s}_0 = s_0 \times L_n = O((\log n)^2) \quad (2.9)$$

where the derivation of this condition arises naturally in the proof of **Theorem 1**. Likewise, it is shown throughout the proofs of the main theorems that $W_n \geq s_0$ is a sufficient condition for guaranteeing that the network has sufficient capacity, in addition to how the bounds imposed on γ are obtained.

KASN Assumptions:

K.1(a): *The strictly stationary process $\{Z_t\}_{t=1}^n = (Y_t, X_t)'$ with compact support is i.i.d. or M -dependent (for $M \in \mathbb{N}$), or*

K.1(b): *β -mixing stationary with exponential decay $\beta(j) \leq \beta_0 e^{-cj}$ for some $\beta_0 > 0$, $c > 0$, or*

K.1(c): *ϕ -mixing (uniform mixing) stationary with $\phi(j) \leq \phi_0 j^{-\xi}$ for some $\phi_0 > 0$, $\xi > 2$.*

K.2(a): *The parameter space Θ is equipped with the pseudometric $\|\cdot\|$, Θ_n (with network parameters p_n , restricting $G_n \geq 2$) is dense in Θ and $\mathcal{F}_d^{m+1} \equiv W_2^{m+1}(\mu)$ is a Sobolev space. Furthermore, the pseudometric $\|\cdot\|$ is equivalent to $K^{1/2}$ on Θ , where $K(\theta_0, \theta) \equiv E_0[l(Z_t, \theta_0) - l(Z_t, \theta)]$ is the Kullback-Leibler-type divergence: there exist constants $0 < c_1 \leq c_2 < \infty$ such that*

$$c_1 K^{1/2}(\theta_0, \theta) \leq \|\theta - \theta_0\| \leq c_2 K^{1/2}(\theta_0, \theta) \quad \forall \theta \in \Theta$$

K.2(b): (Identification) *For all small $\epsilon > 0$ and for some constant $C_1 > 0$:*

$$\sup_{\{\theta \in \Theta_n, \|\theta - \theta_0\| \leq \epsilon\}} \text{Var}[l(Z_t, \theta) - l(Z_t, \theta_0)] \leq C_1 \epsilon^2$$

K.2(c): (QMLE Regularity) *There exists a function $U(\cdot)$ and some $\nu \in (0, 2)$ such that for*

any $\delta > 0$:

$$\sup_{\{\theta \in \Theta_n, \|\theta - \theta_0\| \leq \delta\}} |l(\theta) - l(\theta_0)| \leq \delta^\nu U(Z_t)$$

with $E[|U(Z_t)|^\kappa] \leq C_2$, for some $C_2 > 0$, $\kappa \leq 2$, $\nu\kappa \leq 2$.

Our initial set of assumptions are standard in the literature on sieve extremum estimators. Assumption **K.1** permits data dependence structures beyond the baseline i.i.d. case used to derive the entropy bounds of DNNs. Since M -dependence is generally an unrealistic assumption for economic and financial time-series³⁹, the β -mixing and uniform mixing assumptions (see Section A.2 of the Appendix for full definitions) are necessary for controlling serial dependence in the context of empirical processes. As Chen and Shen (1998) highlight, exponential decay rates in stationary β -mixing processes is satisfied by many nonlinear Markov processes (e.g., a class of nonlinear ARCH models; Doukhan, 1994).

The variance regularity condition in Assumption **K.2(b)** guarantees that θ_0 is identifiable (noting that the sieve space $\mathcal{K}_n$ is compact⁴⁰ by construction) and that $l(\cdot)$ is well-behaved in a neighborhood surrounding it. Correspondingly, the smoothness of the likelihood is dictated by the parameter ν in Assumption **K.2(c)** (where $\nu = 1$ for the quadratic loss function), thus ensuring that the empirical process remains well-behaved as the sieve becomes increasingly fine as $n \rightarrow \infty$.

KASN Assumptions (continued):

K.3(a): (Sparse Compositional Structure) *The target function $f_0 \in \mathcal{F}_d^{m+1} \equiv W_2^{m+1}(\mu)$, $f_0 :$*

³⁹To see why this is the case, consider a Moving Average process of dependence order M (denoted $MA(M)$) with a sequence of i.i.d. shocks ϵ_t defined by $X_t = \epsilon_t + \theta_1\epsilon_{t-1} + \theta_2\epsilon_{t-2} + \dots + \theta_n\epsilon_{t-M}$. For a finite M , this implies that the autocorrelation of this process is zero for all lags greater than M , thus ruling out the long-term persistence common in macroeconomic shocks, to give an example.

⁴⁰We note that $\hat{\theta}_n$ is the *exact* sieve extremum estimate; see Chen (2007) for an expanded discussion in the context of ill-posed problems.

$[0, 1]^d \rightarrow \mathbb{R}$ takes on the following sparse compositional form, where $s_0 \ll 2d + 1$:

$$f_0(X_t) = \sum_{i=1}^{s_0} \Phi_i \left(\sum_{j \in \mathcal{J}_i} \phi_{i,j}(X_{t,j}) \right)$$

with $|\mathcal{J}_i| \leq J_{\max} < \infty$, $J_{\max} \ll d$, and $d = O(n^{\zeta_d})$ for some $\zeta_d > 0$.

K.3(b): (Sparsity Growth) For the $s_n \geq \bar{s}_0$ corresponding to the best approximation $\pi_n \theta_0$, where $\bar{s}_0$ is as defined in (2.9):

i. $s_0 = O(1)$, or

ii. $s_0 = O(\log n)$

K.4(a): Each univariate edge function $\phi_{i,j}, \Phi_i$ in the representation of f_0 belongs to the Sobolev space $W_2^{m+1}(\mathcal{I}_\epsilon)$ with uniformly bounded norm $B < \infty$.

K.4(b): (Lipschitz Continuity) The univariate edge functions $\phi_{i,j}, \Phi_i$ and their derivatives up to order $m + 1$ are Lipschitz continuous, with uniform constant $L < \infty$.

K.4(c): (Compositional Regularity) Each outer function Φ_i in the sparse compositional representation of f_0 admits a depth- $(L_n - 1)$ univariate compositional representation:

$$\Phi_i = \psi_{i,L_n} \circ \psi_{i,L_n-1} \circ \cdots \circ \psi_{i,2}$$

where each $\psi_{i,l} : \mathbb{R} \rightarrow \mathbb{R}$ belongs to $W_2^{m+1}(\mathcal{I}_\epsilon)$ with a uniformly bounded norm and satisfies Assumption **K.4(b)** with Lipschitz constant $L_\psi \leq 1$.

The second set of assumptions focuses on the structural properties of the target function and the regularity of its components. Assumption **K.3(a)** formalizes the sparse compositional structure motivated by the KART (see **Definition 1**) that allows the KAN sieve extremum to bypass the CoD. An interpretable set of primitive conditions under which **K.3(a)** naturally arises is the latent factor model:

$$Y_t = g_0(F_t) + \epsilon_t, \quad F_t = h(X_{t,1}, \dots, X_{t,S}), \quad S \ll d \quad (2.10)$$

where $F_t \in \mathbb{R}^S$ is a low-dimensional latent factor depending on only S covariates and ϵ_t satisfies $\mathbb{E}[\epsilon_t|X_t] = 0$. The conditional expectation $f_0(X_t) = \mathbb{E}[Y_t|X_t] = g_0(F_t)$ then depends on X_t through the subset of cardinality S , so that by the KART, f_0 admits a sparse compositional representation with an active edge count of $O(S)$ rather than $O(d)$. From this perspective, the sparsity level $s_0 \asymp \log n$ is therefore directly interpretable as controlling the number of (active) covariates entering F_t , with the set of relevant factors slowly growing as n increases. This is consistent with the approximate factor model of Stock and Watson (2002), where their framework accommodates $S \rightarrow \infty$ as $n \rightarrow \infty$. Crucially, **K.3(a)** is no more restrictive than the sparsity assumptions commonly found in the econometrics literature: beyond sparse linear regression⁴¹ and sparse additive models (discussed earlier in the context of sparse GAMs), multiple-index models impose $f_0(X_t) = G(X_t'\beta_1, \dots, X_t'\beta_S)$ with $S \ll d$. This corresponds to a KASN where the inner activation functions reduce to linear projections $\phi_j(X_{t,j}) = \beta_j X_{t,j}$ and the outer aggregation (with respect to the unknown function G , as in Ichimura and Lee, 1991) remains nonparametric. The KASN generalizes this by making both components fully nonparametric while permitting deeper compositional representations.

Relevant to the controlled growth of the metric entropy, Assumption **K.3(b)** constrains s_0 to be either constant or at most logarithmic in the sample size. This is consistent with the KASN parameters, as the architecture size (governed by W_n and L_n) also grow logarithmically in n . As emphasized throughout this framework, the expressiveness of KANs is uniquely tied to the grid resolution (G_n). A slowly growing architecture size facilitates the freedom⁴² to select a sufficiently large grid (through $\frac{1}{4k} < \gamma < \frac{1}{2}$) and B-spline coefficient constraint (Δ_n , through hyperparameter $\zeta_\delta > 0$). Since the metric entropy bound in **Theorem 2** is established with respect to the data-driven s_n , the condition $s_n \geq \bar{s}_0$ ensures the sieve space contains a valid approximation to f_0 (provided a sufficiently large Δ_n) without

⁴¹Where sparsity is imposed on the vector of coefficients and the model is approximated by a shallow KASN with linear activation functions.

⁴²Which in general, is crucial for network hyperparameters selected via CV.

requiring knowledge of $\bar{s}_0$.

Assumptions **K.4(a)** and **K.4(b)** are high-level conditions, analogous to assumptions commonly found in deep MLP frameworks, ones that impose the necessary regularity on the univariate edge functions. The Lipschitz continuity of the derivatives further ensures that the sieve extremum objective is stable⁴³. Lastly, Assumption **K.4(c)** is a formalization of the structural assumptions implicit in KAN architectures generalizing the KART to deep architectures (Liu et al., 2024), extended to Lipschitz continuity⁴⁴. This assumption reflects the structural intuition motivated by the literature on multiple-index model, where the inner index functions are linear (and therefore, trivially regular in general) while the outer aggregation G captures the complex nonlinear dependence. **K.4(c)** generalizes this by allowing nonparametric inner functions while imposing the required regularity on the outer compositional structure Φ_i to ensure compatibility with the entropy bound established in **Theorem 2**. Settings where this assumption is most credible involve individual covariate transformations that are smooth but interactions that are complex, such as the factor model structure motivating Assumption **K.3(a)**. The main advantage of our deep architecture (and overparameterization in general) is in the use of additional layers to achieve more efficient representations of the outer functions Φ_i . As a result, our estimator benefits from the improved performance of deeper representations (as discussed in Liu et al. 2024 in comparison to shallow KANs) and the metric entropy growth is controlled via sparsity (through the $s_n \approx \bar{s}_0$ active edges across the L_n layers). This is a key focus in our Monte Carlo simulations under DGPs that involve complex, compositional functions.

With respect to selecting a suitable width such that $W_n \geq s_0$, the use of holdout validation or block-based CV is both practical and consistent with existing deep learning approaches. In theory, overparameterization in any of the parameters p_n does not harm the model per-

⁴³Aiding in the mitigation of the aforementioned exploding gradients problem, as well as functional oscillations that unconstrained KANs are potentially prone to.

⁴⁴Where the constraint $L_\psi \leq 1$ can be loosened to $L_\psi \leq C_\phi < \infty$. This results in a tradeoff that introduces an additive term in the lower bound of γ , one that asymptotically vanishes as $n \rightarrow \infty$. Simplifying the analysis, we proceed with $L_\psi \leq 1$ and show how the aforementioned modified bound arises at the cost of loosening this.

formance due to the adaptability of our sieve space and regularization approach. We verify this using our sparsity ratio (defined in terms of the Δ_n/r_n threshold for active edges) in the simulation analysis across many DGPs, defining $W_n = C_w \log n$ for some constant $C_w > 0$ and evaluating a wide range of potential values. In the case of $s_0 = O(1)$, we simply let $W_n = C_w$ and focus our attention on the case above in proofs involving the metric entropy of this KAN architecture. The requirement that the width of the KASN grows at a rate equivalent to that of s_0 arises from our sparse compositional structure on $f_0 \in \mathcal{F}_d^{m+1}$ (Assumption **K.3(a)**). This ensures that the network possesses enough potential univariate functions to approximate the active edge functions in θ_0 , even as $n \rightarrow \infty$. For simplicity and consistency with standard approaches, we set $L_n = \lfloor \log n \rfloor$ and use the stochastic optimizer Adam (Kingma and Ba, 2014) in practice (see Appendix [A.3](#)).

Under these assumptions, the KASN extremum estimator delivers the convergence rate: $\|\hat{\theta}_n - \theta_0\| = o_P(n^{-1/4})$. We note that while this estimator does not suffer from the CoD as demonstrated in the proofs, d is naturally still relevant to our network parameters through s_0 and the subset $\mathcal{J}_i$. This is despite how estimation methods relying on splines are notorious for failing to produce precise estimates in high-dimensional settings due to the CoD (Chen, 2007). Although there are existing spline estimation approaches that attempt to exploit functional sparsity in high-dimensional environments⁴⁵ similar to our proposed KAN sieve estimator, these methods are likely to perform poorly on the truly compositional structures (i.e., where imposing additivity is insufficient) found in economic and financial applications.

3 Theoretical Results

To simplify the notation, we assume $\theta_0 \equiv f_0 \in \Theta$ and denote the sieve space as $\mathcal{K}_n \equiv \Theta_n$ (i.e., $A \subset \mathbb{R}^a$, the finite-dimensional parameter space, is an empty set) for **Theorems 1-3**. In other words, the estimator is defined as $\hat{\theta}_n = \arg \sup_{\theta \in \mathcal{K}_n} \mathcal{L}_n(\theta)$ with $\hat{\theta}_n \in \mathcal{K}_n$ by construction. Proofs of each of the following theorems can be found in Section [A.1](#) of the

⁴⁵e.g., Meier et al. (2009).

Appendix. Verification of the conditions from Chen and Shen (1998) required for establishing the asymptotic normality of plug-in estimators (**Theorem 4**) follows after the main proofs in Section A.2. Unless stated otherwise, let the network parameters correspond to the conditions outlined in (2.7)-(2.9). In practice, we make use of cubic B-spline activations ($k = 4$) across all simulations and empirical applications, however the following theoretical results hold for all $k \geq m + 1$.

3.1 Rates of Convergence

Theorem 1 (Approximation Error). *Under Assumptions **K.1-K.4** with $\bar{s}_0 = s_0 \times L_n$ and given B-splines of order $k \geq m + 1$ with uniform knots, there exists a sieve approximation $\pi_n \theta_0 \in \mathcal{K}_n$ such that:*

$$\|\pi_n \theta_0 - \theta_0\| \leq C_{\bar{s}_0} G_n^{-k} + C'_{\bar{s}_0} O(n^{-1/2}) = o(n^{-1/4}),$$

where $C_{\bar{s}_0} > 0$ is a constant depending on $\bar{s}_0$, Lipschitz constants (B, L) and spline order k .

The proof of **Theorem 1** relies on the sparse compositional structure imposed on the target function and the smooth univariate edge functions (Assumptions **K.3** and **K.4**). Using results from spline approximation theory (DeVore and Lorentz, 1993), we show that each edge function is approximated with error $O(G_n^{-k})$. The resulting approximation function of the KASN is independent of d , a key result for mitigating the CoD. We note that due to the results from approximation theory applied in the proof of this theorem, we cannot apply the adaptive grid approach of Liu et al. (2024). However, due to our optimization approach that relies on CDF normalization, this is not problematic. As discussed earlier, freedom to choose a sufficiently large $\frac{1}{4k} < \gamma < \frac{1}{2}$ in addition to the constraint Δ_n on the spline coefficients leads to a similar result in practice.

Corollary 1. *Under the same conditions as **Theorem 1** with $k = 4$ and additionally*

assuming that each outer function Φ_i has bounded derivatives $\|\Phi_i^{(r)}\|_\infty \leq B_r < \infty$ for $r = 1, \dots, m+1$:

$$\|\pi_n \theta_0 - \theta_0\|_{W_2^{m+1}(\mu)} \leq C_{\bar{s}_0} G_n^{-1} + C'_{\bar{s}_0} O(n^{-1/2}) = o(n^{-1/4}),$$

with $\gamma > \frac{1}{4}$ sufficient for all $m \leq 2$.

Remark: The Sobolev norm bound in **Corollary 1** is directly relevant to Example 2, where the functional of interest is the average derivative $\Gamma(\theta_0) = \mathbb{E}[\partial\theta_0(X_t)/\partial x_j]$, which depends on θ_0 through its first-order partial derivatives. Control of the approximation error in the $W_2^{m+1}(\mu)$ norm is therefore necessary for consistent estimation of $\Gamma(\theta_0)$, and the bound $O(G_n^{-1})$ under $k = 4$ is sufficient for the $o_P(n^{-1/4})$ rate required by **Theorem 4**. Furthermore, in both **Theorem 1** and **Corollary 1**, the deterministic terms dominate the stochastic $O(n^{-1/2})$ components introduced through the normalization of the inputs from (2.4) in the approximation error bounds.

Theorem 2 (Entropy Bound). *Under Assumptions **K.1-K.4**, and given the sieve space $\mathcal{K}_n$ with $s_n \geq \bar{s}_0$:*

$$\log \mathcal{N}(\epsilon, \mathcal{K}_n, \|\cdot\|_{L_2(\mu)}) \leq C_1 s_n \log \left(\frac{r_n}{s_n} \right) + C_2 s_n G_n \log \left(\frac{\Delta_n}{\epsilon} \right) + o(1),$$

where $s_n \geq \bar{s}_0 = s_0 \times L_n$ is the active edge count satisfying $s_n = O(\bar{s}_0)$.

In this proof, the metric entropy bound in terms of covering numbers is broken down into two components. The first, $s_n \log(r_n/s_n)$, is a combinatorial term (by Lemma 17.5.1 of Cover and Thomas, 1991; where $p = s_n/r_n$. A similar argument appears in Raskutti et al., 2011) arising from the active edge selection which is central to our sparse compositional form assumption. The B-spline complexity term $s_n G_n \log(\Delta_n/\epsilon)$ arises via the $\ell_{1,2}$ -constraint imposed on the spline coefficients (by Corollary 4.2.11 of Vershynin, 2018), which encourages sparsity across the r_n potential edges. This highlights the role of Δ_n in controlling the overall complexity of $\mathcal{K}_n$ in the KASN architecture and the dependence on effective parameters

$(s_n G_n)$ only, crucial for high-dimensional estimation.

The $o(1)$ term accounts for the residual connection mapping h introduced in the definition of $\mathcal{K}_n$. Since h is a fixed, non-parameterized element, its covering number is unity, thus contributing a null term to the metric entropy $\log \mathcal{N}(\epsilon, \mathcal{K}_n, \|\cdot\|_{L_2(\mu)})$. We note that the bound established here is conservative in nature, since it treats s_n as if each active edge is allocated the full coefficient constraint Δ_n , thereby simplifying the analysis. In order to apply the main result (Theorem 1) of Chen and Shen (1998) for the sake of establishing the asymptotic convergence rate of the KASN estimator, we require the $L_2(\mu)$ -bracketing entropy established in **Corollary 2**.

Corollary 2. *Under Assumptions **K.1-K.4**, the $L_2(\mu)$ -bracketing entropy of $\mathcal{K}_n$ satisfies:*

$$\begin{aligned} \log \mathcal{N}_{[]}(\epsilon, \mathcal{K}_n, \|\cdot\|_{L_2(\mu)}) &\leq \log \mathcal{N}(\epsilon/2, \mathcal{K}_n, \|\cdot\|_{L_2(\mu)}) \\ &\leq C_1 s_n \log \left(\frac{r_n}{s_n} \right) + C_2 s_n G_n \log \left(\frac{2\Delta_n}{\epsilon} \right) + o(1) \end{aligned}$$

Theorem 3 (Estimation Error). *Under Assumptions **K.1-K.4**, given the sieve space $\mathcal{K}_n$ and letting $\frac{1}{4k} < \gamma < \frac{1}{2}$:*

$$\|\hat{\theta}_n - \pi_n \theta_0\| = o_P(n^{-1/4})$$

The proof of **Theorem 3** involves splitting the entropy integral at $\epsilon_0 = n^{-1}$ and showing that each term is $o_P(n^{-1/4})$. Obtaining convergence rates in the Sobolev norm would lead to a condition that $m > d$ as we show in the Appendix, which would re-introduce the CoD. Instead, the estimation error achieves a convergence rate of $o_P(n^{-1/4})$ which holds uniformly over $d = O(n^{\zeta_d})$.

Corollary 3. *Under Assumptions **K.1-K.4**, given the sieve space $\mathcal{K}_n$ and letting $\frac{1}{4k} < \gamma < \frac{1}{2}$, the KASN extremum estimator achieves $\|\hat{\theta}_n - \theta_0\| = o_P(n^{-1/4})$.*

To see this, note that by the triangle inequality:

$$\begin{aligned}\|\hat{\theta}_n - \theta_0\| &\leq \|\hat{\theta}_n - \pi_n \theta_0\| + \|\pi_n \theta_0 - \theta_0\| \\ \|\hat{\theta}_n - \theta_0\| &= o_P(n^{-1/4}) + o(n^{-1/4}) = o_P(n^{-1/4})\end{aligned}$$

Remark: By letting $p_n \rightarrow \infty$ and $\Delta_n \rightarrow \infty$ as $n \rightarrow \infty$, we obtain a sequence of increasingly flexible network architectures; making this analogous to the universal approximator result from White (1990) for MLPs, now extended to KANs under functional sparsity. For sparse compositional target functions, KASNs represent a class of *connectionist sieve networks* with multiple hidden layers that guarantees a dimension-free convergence rate, the first of its kind.

While Gouriéroux and Monfort (2025) correctly note that the KART is non-unique at the component level (with respect to $\phi_{i,j}$ and Φ_i), this does not affect the identification of the target function itself. The KASN framework formalizes identification of θ_0 as the exact maximizer of the population criterion rather than identification of the individual component functions. The compact sieve space $\mathcal{K}_n$ and the identification condition in Assumption **K.2(b)** guarantee the existence of a measurable maximizer $\hat{\theta}_n$ of the empirical criterion, which is sufficient for the convergence results in **Theorems 1–3** and the asymptotic normality result (independent of the underlying architecture) in **Theorem 4**. Beyond measurability, the identification issues they raise surrounding the CoD are resolved directly: **Corollary 3** establishes that the convergence rate $o_P(n^{-1/4})$ holds uniformly over $d = O(n^{\zeta_d})$, rendering dimensional scaling irrelevant to the estimation problem.

In **Theorem 4**, we establish the asymptotic normality of plug-in estimators using KASNs. Following Chen and Shen (1998, Theorem 2), we consider a linear functional $\Gamma : \Theta \rightarrow \mathbb{R}$ with $v^* \in \bar{V}$ (by the Riesz representation theorem) satisfying $\Gamma'_{\theta_0} = \langle \theta - \theta_0, v^* \rangle$ for all $\theta \in \Theta$. The existence of v^* , the Gateaux derivative of Γ , ensures that the functional is bounded (where $\bar{V}$ denotes the completion of the space spanned by $\Theta - \{\theta_0\}$ and the inner product $\langle \cdot, \cdot \rangle$ is induced by $\|\cdot\|$) and identifies the direction in the sieve space that characterizes the

asymptotic variability of the estimator. We expand on these definitions and list the relevant conditions from Chen and Shen (1998) in the Appendix, before verifying each condition for the KASN.

Theorem 4 (Asymptotic Normality). *Under Assumptions **K.1-K.4** and **B.1-B.5** (Appendix A.2), the plug-in KASN estimator satisfies:*

$$\sqrt{n}\left(\Gamma(\hat{\theta}_n) - \Gamma(\theta_0)\right) \rightsquigarrow N(0, \sigma_*^2)$$

$$\sigma_*^2 = \lim_{n \rightarrow \infty} n^{-1} \text{Var}\left(\sum_{t=1}^n l'_{\theta_0}[v^*, Z_t]\right),$$

where $\sigma_*^2 = \text{Var}(l'_{\theta_0}[v^*, Z_1]) + 2 \sum_{j=2}^{\infty} \text{Cov}(l'_{\theta_0}[v^*, Z_1], l'_{\theta_0}[v^*, Z_j])$.

In ultra-high-dimensional settings ($d \gg n$), estimates of this linear component (e.g., the parameter of interest in the partially linear regression model introduced in Example 1) are likely to suffer from regularization bias, leading us to consider an alternative approach centered around the DML framework (Chernozhukov et al., 2018) designed to address this.

3.2 Double/Debiased Machine Learning

In this section, we show how our KASN extremum estimator can be applied within the DML framework of Chernozhukov et al. (2018) to obtain $\sqrt{n}$ -consistent and asymptotic normal estimates of linear parameters of interest. A similar DML application appears in Farrell et al. (2025), although the convergence rate of the ReLU DNN they propose explicitly depends on the heterogeneity dimension (d_c). In addition to the presence of the CoD, their near-minimax $O(n^{-m/(m+d_c)} \log^8 n)$ convergence rate depends on the smoothness of the target function; highlighting the tradeoff arising from the highly useful economic interpretability of their proposed estimator. To the best of our knowledge, the KASN is the first and only network architecture since Chen and White (1999) capable of delivering a dimension-free convergence rate of $o_P(n^{-1/4})$.

Chernozhukov et al. (2018) address how the naive application of machine learning algorithms in semiparametric models with low-dimensional causal parameters can lead to biased estimates. This arises from the regularization methods commonly employed in these models for nuisance function estimation, which perform well by reducing variance (overfitting) in a manner consistent with the standard variance-bias tradeoff. As a result of this overfitting and regularization bias, estimates of the causal parameter obtained by using the fitted nuisance function as a plug-in estimator will suffer from a heavy degree of bias. Furthermore, they demonstrate how the Donsker conditions typically imposed in semiparametric models to control complexity break down in ultra-high-dimensional settings. The DML framework resolves these issues through two key ingredients: Neyman orthogonality and sample splitting.

We consider the partially linear regression model⁴⁶ (Robinson, 1988) in our simulations and assume $Z_i = \{Y_i, X_i\} = \{Y_i, D_i, W_i\}$ is i.i.d. for this application. This model is defined in terms of the following equations:

$$Y_i = D_i\alpha_0 + g_0(W_i) + U_i, \quad E[U_i|W_i, D_i] = 0 \quad (3.1)$$

$$D_i = m_0(W_i) + V_i, \quad E[V_i|W_i] = 0 \quad (3.2)$$

where $V_i = D_i - m_0(W_i)$ and the sample splitting procedure (in the case of two folds) entails estimating m_0 and g_0 using half of the dataset, before forming residuals on the other half. In this model, D_i represents the treatment variable (or more generally, the primary regressor of interest) with a linear causal effect α_0 estimated after controlling for the confounding effects of W_i . Neyman orthogonality is the condition that ensures moment conditions defining α_0 are insensitive to perturbations⁴⁷ in the nuisance functions $\eta_0 = (g_0, m_0) \in \mathcal{T}$ (where $\mathcal{T}$ is a convex subset of a normed vector space. For simplicity, let $\mathcal{T} \equiv \Theta$).

⁴⁶Which is the leading example in Chernozhukov et al. (2018); see our partition of X_i defined earlier.

⁴⁷With respect to errors in estimating both the outcome model and the treatment/propensity model.

Definition 2 (Neyman Orthogonality). *The score $\psi(\cdot; \cdot)$ satisfies Neyman orthogonality if:*

$$\partial_\eta E[\psi(Z; \alpha_0, \eta_0)][\eta - \eta_0] = 0 \quad \text{for all } \eta \in \mathcal{T} \quad (\text{DML.1})$$

The main advantage of our procedure estimating nuisance functions using KASNs over widely used machine learning methods (e.g., Lasso, random forests) is how it is capable of capturing complex compositional structures⁴⁸ in the data-generating process (DGP). In their empirical example, Chernozhukov et al. (2018, Section 6.1) discuss how “Deep Learning methods” were experimented with but that the results were omitted due to computational and stability issues encountered. We observe no such issues across all of our Monte Carlo simulations and compare the results against those of a Lasso model using a penalty λ_L selected via cross-validation. By Theorem 4.1 of Chernozhukov et al. (2018), the following result holds when σ^2 is replaced with $\hat{\sigma}^2$.

Proposition 1 (DML Estimator). *Under Assumptions **K.1(a)**-**K.4**, the DML estimator $\hat{\alpha}$ where $(\hat{g}_n, \hat{m}_n) \in \mathcal{K}_n$ satisfies:*

$$\sigma^{-1} \sqrt{n}(\hat{\alpha} - \alpha_0) \rightsquigarrow N(0, 1), \quad \sigma^2 = \frac{E[V_i^2 U_i^2]}{(E[V_i^2])^2}, \quad (\text{DML.2})$$

where by **Corollary 3**, $\hat{g}_n$ and $\hat{m}_n$ each achieve $o_P(n^{-1/4})$ in $L^2(\mu)$, so that

$$\|\hat{g}_n - g_0\| \cdot \|\hat{m}_n - m_0\| = o_P(n^{-1/2}),$$

thus satisfying the rate condition of Theorem 4.1 of Chernozhukov et al. (2018). We implement the K -fold cross-fitting procedure of Chernozhukov et al. (2018), estimating $(\hat{g}_n, \hat{m}_n)$ on each complementary fold and forming residuals on the held-out fold.

⁴⁸Nonlinear or otherwise.

4 Monte Carlo Simulations

In this section, we analyze the performance of the KASN in Monte Carlo simulations. We first consider a simple additive nonparametric regression model in a moderately high-dimensional setting. Under a more complex data generating process (DGP), we then evaluate our proposed estimator in a DML application (using the PLR model outlined in Eq. (3.1)-(3.2) and Example 1) against the performance of other ML methods. Both involve running $N = 1000$ total simulations and evaluating the aggregate results, with careful attention paid to the crucial sparsity ratio. This is the ratio of inactive edges relative to the total number of potential edges (r_n), using the Δ_n/r_n threshold for what constitutes an active edge (i.e., learned univariate functions with a magnitude greater than those in a hypothetical fully dense network) defined earlier. We also report results obtained using the sieve network from Chen and White (1999) denoted by “SLFN”, which provides a shallow network model comparison for our KASN.

4.1 Finite-Sample Performance

In our baseline nonparametric regression, we evaluate ability of the KASN to recover a sparse additive signal with a target function dependent on $S = 30$ out of $d = 100$ total covariates and $n = 5000$ observations. The KASN parameters are set consistent with the ones outlined in Section 2.3, where we manually select $\gamma = 0.35$.

$$f_0(X) = \sum_{j=1}^S \sin(X_j) + \epsilon, \quad \epsilon \sim N(0, 0.25) \quad (4.1)$$

As seen in Table 1, the KASN achieves a Test R^2 of 0.78 and a Test RMSE of 1.87, where both metrics are comparable to the performance on the training data and significantly outperforming the SLFN. The gap in performance of these models is indicative of the superior approximation power possessed by the KASN, demonstrating how the learned B-spline edges directly adapt to the smoothness and periodicity of the target function. With respect to

performance on unseen data, we observe minimal overfitting of the KASN (despite our novel and relatively aggressive regularization method) based on the differences between the aforementioned metrics. On the other hand, the SLFN reliant on a fixed activation function suffers from underfitting as it lacks the localized flexibility of the B-spline grid.

Table 1: Nonparametric Regression Results

Metric	KASN	SLFN
Train RMSE	1.66	2.34
Train R^2	0.83	0.66
Test RMSE	1.87	2.39
Test R^2	0.78	0.64
Runtime (s)	2513.88	0.33
Sparsity Ratio (%)	97.61	N/A

The most critical finding in this setting is the effective dimensionality reduction achieved through the Group Lasso penalization. Out of the many r_n potential edges, the KASN achieves a sparsity ratio of 97.61%. This means that the network successfully identified the narrow subset of active edges required to represent the $S = 30$ relevant⁴⁹ covariates, resulting in $s_n = 4339$ active (unpruned) univariate functions. This prevents the overfitting noise common in insufficiently penalized deep networks and is crucial for maintaining a controlled metric entropy as discussed throughout this paper. The relatively higher computational runtime (with an average of 502.8 seconds per fold) is consistent with the tradeoff associated with deeper networks, as the KASN demonstrates significantly higher statistical efficiency. These results verify that even in cases where the true model is additive rather than deeply compositional (e.g., in our next simulation setup), the KASN extremum estimator is robust and highly accurate; with an underlying architecture that is naturally robust to the CoD.

In our DML simulations, we compare the performance of KASNs against Lasso in estimating $\alpha_0 = 1.5$. Across each DGP, we generate $d = 100$ covariates in total with $S = 30$ relevant covariates and set use five folds in the sample splitting procedure. The test set serves

⁴⁹For brevity, we refer to covariates upon which f_0 depends on as ‘relevant’.

an important purpose in making sure our fitted models generalize well to unseen data and we report these loss metrics. We fix the KASN parameter determining the grid size as $\gamma = 0.499$ for all simulations. In each DGP, the true partially linear regression model is generated using the following nonlinear compositional functions and computed orthogonal residuals:

$$m_0(\mathbf{X}) = \sin\left(\sum_{j=1}^S \frac{1}{1 + e^{-X_j}}\right) + 0.1 \sum_{j=1}^S X_j^2 + \epsilon_m, \quad \epsilon_m \sim N(0, 0.25) \quad (4.2)$$

$$g_0(\mathbf{X}) = \frac{1}{1 + e^{-\sum_{j=1}^S \sin(X_j)}} + 0.1 \sum_{j=1}^S |X_j| + \epsilon_g, \quad \epsilon_g \sim N(0, 0.25) \quad (4.3)$$

where $\hat{V} = D - \hat{m}(\mathbf{X})$, and $\hat{U} = Y - \hat{g}(\mathbf{X})$ are used to obtain $\hat{\alpha} = \frac{\sum \hat{V}_i(Y_i - \hat{g})}{\sum \hat{V}_i D_i}$. For our main simulation DGP, we have a KASN with $L_n = 6$ and $W_n = 201$ (KASN #1). We also consider an additional KASN with $W_n = 101$ (KASN #2).

Table 2: DML Simulation Results: ($\alpha_0 = 1.5$)

Metric	KASN #1	KASN #2	SLFN	Lasso
Causal Parameter ($\hat{\alpha}$)				
$\hat{\alpha}$	1.5024	1.4804	1.6660	1.7318
95% CI	[1.4594, 1.5454]	[1.4368, 1.5240]	[1.6324, 1.6996]	[1.7077, 1.7559]
RMSE	0.0657	0.0461	0.1672	0.2323
Nuisance Function Performance				
Train R^2 ($\hat{m}_0 / \hat{g}_0$)	0.80 / 0.82	0.84 / 0.85	0.09 / 0.07	0.18 / 0.16
Test R^2 ($\hat{m}_0 / \hat{g}_0$)	0.39 / 0.45	0.36 / 0.43	-0.04 / -0.06	-0.13 / -0.17
Sparsity Ratio (%) ($\hat{m}_0 / \hat{g}_0$)	92.5 / 92.9	88.8 / 89.0	NA / NA	8.35 / 4.99
Runtime (s)	2302.98	1070.20	0.17	0.04

Across the DML simulations, the KASN demonstrates a strong ability to recover the true causal parameter $\alpha_0 = 1.5$ despite the complex compositional structure of the nuisance functions in the DGP. The wider network (KASN #1 in Table 2) achieves a near-zero bias, yielding $\hat{\alpha} = 1.5024$ while the narrower architecture (KASN #2) yields $\hat{\alpha} = 1.4804$. In contrast, the SLFN and Lasso estimators exhibit substantial upward bias, with estimates of $\hat{\alpha} = 1.666$ and $\hat{\alpha} = 1.7318$, respectively. Despite the relatively strong performance of the SLFN in the earlier nonparametric regression (including its ability to generalize to unseen

data, captured by the Test R^2), it fails to learn the compositional structures of m_0 and g_0 . This highlights how shallow networks generally struggle to decompose complex hierarchical nonlinearities, similar to the the Lasso model which imposes a linear structure: resulting in a failure to eliminate regularization bias even under Neyman Orthogonality and sample-splitting. Conversely, both iterations of the KASN display significant predictive power on the test set; confirming our theory that the learned B-spline activations effectively decomposes compositions into univariate sieve components, whereas the internal structure is unidentifiable to other architectures.

We observe that overparameterization in W_n does not degrade performance. Despite the larger width of KASN #1, it maintains high sparsity ratios, indicating the success of our regularization method combining the Group Lasso with the constraint coefficient Δ_n . As emphasized throughout this paper and in our theoretical results, this structural sparsity is crucial for controlling the KASN metric entropy in high-dimensions while preserving approximation power; thus preventing the explosion in complexity that standard MLPs are prone to. Furthermore, the stability of our estimator across different widths reinforces the robustness of the sieve grid resolution G_n . Even when the width is halved, the KASN extremum estimator maintains strong performance and ability to generalize to unseen data. These results provide empirical evidence that the KASN is a superior ML method for estimating nuisance functions in high-dimensional settings characterized by compositional covariate effects.

5 Empirical Application

We evaluate our proposed KASN extremum estimator in an empirical application featuring a large set of covariates relative to the overall sample size. Rapach and Zhou (2021) demonstrate how principal component analysis (PCA) can be applied to 118 macroeconomic and financial variables in order to identify interpretable risk factors. The resulting components (e.g., housing, yields, credit spreads) are linear combinations of subsets of the aforementioned

variables obtained from the Federal Reserve Economic Data Monthly Database (FRED-MD, 1963-2024), which are then applied to estimate risk premia for a large set of test assets. See the online appendix of Rapach and Zhou (2021) for an outline of the standard transformations applied to the data. We split the $n = 726$ observations using a 0.8 : 0.1 : 0.1 train/validation/test ratio.

In our initial application, we perform a simple forecasting exercise evaluating the ability of the KASN estimator to forecast changes in interest rates. We consider the 1-year constant maturity Treasury yield (GS1) from FRED-MD, constructing a 12-month lag structure over the full sample. For this exercise alone, we do not make use of early stopping (see Section A.3 in the Appendix), with KASN hyperparameters selected consistent with what is outlined in Section ???. Table 3 reports the results, the KASN achieves a low train RMSE, consistent with the overparameterization outlined earlier.

Table 3: Comparative Performance: LASSO, SLFN, and KASN (12-Month Lag, GS1)

Metric	LASSO	SLFN	KASN
Train RMSE	0.4525	0.3559	0.0265
Test R^2	0.1154	-0.2200	0.5266
Sparsity (%)	88.89	N/A	98.32
Training Time (s)	1.40	9.76	117.03
DM Statistic (vs. KASN)	2.0695	2.5543	N/A
p-value (5% sig.)	0.0385	0.0106	N/A

We find that in terms of the out-of-sample generalization in the form of the Test R^2 , the KASN significantly outperforms our two comparison models. The robustness of this result is supported by additional specifications (increasing the size of the test set, evaluating 18-month lags, using FEDFUNDS as the target variable instead of GS1) and these results can be found in the Appendix. Formal comparison is provided by Diebold-Mariano (DM) tests (Diebold and Mariano, 1995) of equal predictive accuracy against the KASN. The DM statistic for both the LASSO (2.070) and the SLFN (2.554) comparison models are rejected at the 5% level, confirming that the out-of-sample advantage of the KASN is statistically

significant and not attributable to sampling variation.

The findings of Rapach and Zhou (2021) suggests the existence of sparse linear relationships between the raw macro-finance variables, which we use (in addition to the aforementioned key sparse components) to evaluate our model against Lasso in a simple nonparametric regression. Given how this is the type of setting that the Lasso model traditionally excels in, this serves as a robustness test for the KASN involving noisy time-series financial data. Selecting $\gamma = 0.499$ and $\zeta_\delta = 0.9$, the KASN parameters are given by: $L_n = 6$, $W_n = 243$, $G_n = 24$ and $\Delta_n = 307.91$.

Table 4: Sparse Macro-Finance Factors

Metric	LASSO	SLFN	KASN
Train RMSE	4.49	3.85	3.94
Train R^2	0.38	0.54	0.52
Test RMSE	4.70	4.53	4.75
Test R^2	0.52	0.56	0.51
Runtime (s)	1.85	58.11	534.46
Sparsity (%)	89.3	N/A	97.7

As shown in Table 4, the KASN achieves a Test R^2 of 0.51, which is nearly identical to the that of the Lasso model. This is of significant as it provides empirical evidence of the robustness of our sieve extremum estimator in high-dimensional settings characterized by (approximate) linearity. In regimes involving linear or additive DGPs, overparameterized MLP architectures generalize poorly (an issue addressed by Schmidt-Hieber, 2020) due to the unconstrained parameter space. Conversely, the KASN architecture effectively contracts or collapses to the linear approximation. Combined with our DML simulation results, this displays robustness to both extremes (with respect to the target function complexity) in its ability to generalize. Given this demonstrated robustness of our proposed estimator, the tradeoff resulting in increased computation time is justified. The Lasso model produces non-zero coefficient estimates for 12 covariates, while the KASN demonstrates a sparsity ratio of 97.7%. By pruning a significant proportion of redundant edges, the KASN minimizes

the risk of capturing spurious nonlinearities; which is crucial for preserving consistency in finite-samples, especially when n is small relative to the total number of covariates as it is in this application.

6 Conclusion

This paper introduces the KASN estimator, a novel sieve extremum estimator that integrates the adaptive functional flexibility of KANs with the method of sieves. Our framework addresses a critical gap in the deep learning literature by providing an architecture that is innately robust to the CoD under structural sparsity and delivers a convergence rate of $o_P(n^{-1/4})$. This is achieved through the use of learnable B-spline activation functions and a novel regularization method involving the Group Lasso.

The established theoretical results demonstrate that the KASN maintains a controlled entropy under a sparse compositional assumption imposed on the target function. This enables consistent estimation in high-dimensional settings, where the number of covariates is large relative to the sample size. Furthermore, KASN estimator converges quickly enough to deliver asymptotic normal plug-in estimators. Unlike standard MLPs, which often suffer from overparameterization and poor generalization in sparse environments, the KASN effectively prunes redundant univariate functions to identify and recover the underlying signal of the target function. Our Monte Carlo simulations and empirical application further substantiates these theoretical claims, with the KASN consistently producing a high ratio of inactive edges relative to the total potential edges (across a wide range of architectural sizes, dictated by network lengths and widths). The KASN performs especially well in DML simulations involving highly complex compositional nuisance functions relative to other ML methods evaluated. An extension integrating a sample-splitting method suitable for time-series data is left to future work, although this is a very promising finding with respect to the rapidly-growing literature on DML.

In summary, the KASN estimator introduced in this paper represents a tractable and statistically efficient alternative for nonparametric estimation. By bridging the gap between GAMs and deep compositional networks, it provides a robust and flexible approach to approximation that rigorously controls the complexity of the function space in high-dimensional settings.

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A Appendix

A.1 Proofs of Main Results

Table 5: KASN Architecture Parameter Definitions and Growth Rates

Parameter	Condition
Number of Layers	$L_n \asymp \log n$
Width	$W_n \asymp s_0 \log n$
Grid Size	$G_n \asymp n^\gamma, \frac{1}{4k} < \gamma < \frac{1}{2}$
Coefficient Radius	$\Delta_n \asymp n^{\zeta_\delta}, \quad \zeta_\delta > 0$
Total Active Edges	$s_n \geq \bar{s}_0 = s_0 \times L_n$
Total Potential Edges	$r_n = L_n W_n^2$
Total Parameters	$p_n \approx r_n G_n$

Proof of Theorem 1: Under **K.3(a)**, θ_0 admits a sparse compositional structure with inner edges given by $\phi_{i,j}$ for $j \in \mathcal{J}_i$ (noting that $\forall j \notin \mathcal{J}_i, \phi_{i,j} \equiv 0$) which are sufficiently smooth (**K.4(a)**) to be approximated by B-splines of order $k \geq m + 1$. Since $\phi_{i,j}$ depends only on $X_{t,j}$, the $L_2(\mu) \equiv L_2(\tilde{\mu})$ (see the definition in Section 2 in the main text) norm reduces to the marginal norm $\|\cdot\|_{L_2(\mu_j)}$. By **K.4(a)**, $\|\phi_{i,j}^{(k)}\|_\infty \leq B < \infty$, therefore by Theorem 7.3 of DeVore and Lorentz (1993) with uniform knots of mesh $\delta_T = O(G_n^{-1})$, the spline approximation error satisfies $\|\phi_{i,j} - \pi_n \phi_{i,j}\|_{L_\infty(\mathcal{I}_\epsilon)} \leq C_1 B G_n^{-k}$. Since $\tilde{\mu}_j$ is a probability measure supported on $[0, 1] \subset \mathcal{I}_\epsilon$, the L_∞ bound on the padded domain $\mathcal{I}_\epsilon$ implies the $L_2(\tilde{\mu}_j)$ bound directly:

$$\begin{aligned}
\|\phi_{i,j} - \pi_n \phi_{i,j}\|_{L_2(\mu)} &= \|\phi_{i,j} - \pi_n \phi_{i,j}\|_{L_2(\mu_j)} \\
&\leq \|\phi_{i,j} - \pi_n \phi_{i,j}\|_{L_\infty(\mathcal{I}_\epsilon)} \\
&\leq C_1 B G_n^{-k},
\end{aligned} \tag{T1.1}$$

where $C_1 > 0$ depends only on k .

Define $u_i = \sum_{j \in \mathcal{J}_i} \phi_{i,j}(X_{t,j})$ and $v_i = \sum_{j \in \mathcal{J}_i} \pi_n \phi_{i,j}(X_{t,j})$. By the triangle inequality and

(T1.1):

$$\begin{aligned}\|u_i - v_i\| &\leq \sum_{j \in \mathcal{J}_i} \|\phi_{i,j} - \pi_n \phi_{i,j}\|_{L_2(\mu)} \\ &\leq |\mathcal{J}_i| C_1 B G_n^{-k} \leq C_2 B G_n^{-k},\end{aligned}\tag{T1.2}$$

absorbing $|\mathcal{J}_i|$ into C_2 since $|\mathcal{J}_i| \ll d$ is bounded under **K.3(a)**. Applying **K.4(b)** with Lipschitz constant L :

$$\begin{aligned}\|\Phi_i(v_i) - \Phi_i(u_i)\| &\leq L \|v_i - u_i\| \\ &\leq C_2 L B G_n^{-k}\end{aligned}\tag{T1.3}$$

Under **K.4(c)**, $\Phi_i = \psi_{i,L_n} \circ \dots \circ \psi_{i,2}$ with each univariate $\psi_{i,l} \in W_2^{m+1}(\mathcal{I}_\epsilon)$. The KASN approximates each $\psi_{i,l}$ by $\pi_n^{(l)} \psi_{i,l}$, a layer- l B-spline. Applying the following the decomposition:

$$\begin{aligned}\Phi_i - \pi_n \Phi_i &= \sum_{l=2}^{L_n} \left[\psi_{i,L_n} \circ \dots \circ \psi_{i,l+1} \circ \psi_{i,l} \circ \pi_n^{(l-1)} \psi_{i,l-1} \circ \dots \circ \pi_n^{(2)} \psi_{i,2} \right] \\ &\quad - \left[\psi_{i,L_n} \circ \dots \circ \psi_{i,l+1} \circ \pi_n^{(l)} \psi_{i,l} \circ \pi_n^{(l-1)} \psi_{i,l-1} \circ \dots \circ \pi_n^{(2)} \psi_{i,2} \right]\end{aligned}$$

Applying **K.4(c)** to propagate through layers $l+1, \dots, L_n$ with uniform Lipschitz constant $L_\phi \leq 1$, we have:

$$\|\Phi_i - \pi_n \Phi_i\| \leq \sum_{l=2}^{L_n} L_\phi^{L_n-l} \|\psi_{i,l} - \pi_n^{(l)} \psi_{i,l}\| \tag{T1.4}$$

Applying (T1.1) to each univariate $\psi_{i,l}$:

$$\|\psi_{i,l} - \pi_n^{(l)} \psi_{i,l}\| \leq C_1 B G_n^{-k} \tag{T1.5}$$

Substituting into (T1.4):

$$\begin{aligned}\|\Phi_i - \pi_n \Phi_i\| &\leq C_1 B G_n^{-k} \sum_{l=2}^{L_n} L_\psi^{L_n-l} \\ &= C_1 B G_n^{-k} \cdot \frac{L_\psi^{L_n-1} - 1}{L_\psi - 1},\end{aligned}\tag{T1.6}$$

where since $L_\psi \leq 1$ by **K.4(c)**: if $L_\psi < 1$, the geometric sum satisfies

$$\sum_{l=2}^{L_n} L_\psi^{L_n-l} = (1 - L_\psi^{L_n-1}) / (1 - L_\psi) \leq 1 / (1 - L_\psi) < \infty,$$

and if $L_\psi = 1$, the sum equals $L_n - 1 = O(\log n)$. In both cases the sum is $O(\log n)$, absorbed into the constant $C_3 > 0$.

$$\|\Phi_i - \pi_n \Phi_i\| \leq C_3 B G_n^{-k}\tag{T1.7}$$

The probability measure of the inputs μ is approximated by $\tilde{\mu}$ via the empirical distribution function. By van der Vaart (1998) Theorem 19.3 and by an argument identical to (T1.1), $\|\hat{F}_n^j - F^j\|_\infty = O(n^{-1/2})$ for each marginal CDF F_n^j , uniformly over $d = O(n^{\zeta_d})$, contributing:

$$C'_{\bar{s}_0} O(n^{-1/2})\tag{T1.8}$$

Summing (T1.3) and (T1.7) over $i = 1, \dots, s_0$ and adding (T1.8):

$$\begin{aligned}\|\pi_n \theta_0 - \theta_0\| &\leq s_0 (C_2 L + C_3) B G_n^{-k} + C'_{\bar{s}_0} O(n^{-1/2}) \\ &= C_{\bar{s}_0} G_n^{-k} + C'_{\bar{s}_0} O(n^{-1/2}),\end{aligned}\tag{T1.9}$$

where $C_{\bar{s}_0} \equiv s_0 (C_2 L + C_3) B = O(\bar{s}_0)$ since $C_3 = O(\log n)$ from earlier and $\bar{s}_0 = s_0 L_n =$

$O((\log n)^2)$. Under $G_n \asymp n^\gamma$ and $\bar{s}_0 = O((\log n)^2)$, we have:

$$C_{\bar{s}_0} G_n^{-k} = O((\log n)^2 \cdot n^{-k\gamma}),$$

and $-k\gamma < -\frac{1}{4} \iff \gamma > \frac{1}{4k}$ is sufficient for the resulting $o(n^{-1/4})$ rate. $\square$

Proof of Corollary 1: For each $\phi_{i,j} \in W_2^{m+1}(\mathcal{I}_\epsilon)$ and $0 \leq l \leq m$, by Schumaker (2007) Theorem 12.8 with $k = 4 \geq m + 1$ (for $m \leq 3$):

$$\|\phi_{i,j}^{(l)} - (\pi_n \phi_{i,j})^{(l)}\|_{L^2(\mu)} \leq C_k B G_n^{-(m+1-l)}, \quad (\text{C1.1})$$

where under the Sobolev norm:

$$\begin{aligned} \|\phi_{i,j} - \pi_n \phi_{i,j}\|_{W_2^{m+1}(\mu)}^2 &= \sum_{l=0}^{m+1} \|(\phi_{i,j} - \pi_n \phi_{i,j})^{(l)}\|_{L^2(\mu)}^2 \\ &\leq C_k^2 B^2 G_n^{-2} \sum_{r=0}^m G_n^{-2r}, \end{aligned} \quad (\text{C1.2})$$

by re-indexing $r = m - l$. Since $G_n \geq 2$ under **K.2(a)**, the geometric series satisfies $\sum_{r=0}^m G_n^{-2r} \leq \frac{1}{1-G_n^{-2}} \leq \frac{4}{3}$ uniformly given $m \leq 3$ and $G_n \geq 2$. Therefore:

$$\|\phi_{i,j} - \pi_n \phi_{i,j}\|_{W_2^{m+1}(\mu)} \leq \sqrt{\frac{4}{3}} C_k B G_n^{-1} \quad (\text{C1.3})$$

Given u_i and v_i defined earlier:

$$\begin{aligned} \|u_i - v_i\|_{W_2^m}^2 &\leq |\mathcal{J}_i| \sum_{j \in \mathcal{J}_i} \sum_{l=0}^m \|(\phi_{i,j} - \pi_n \phi_{i,j})^{(l)}\|_{L^2(\mu)}^2 \\ &\leq \frac{4}{3} |\mathcal{J}_i|^2 C_k^2 B^2 G_n^{-2}, \end{aligned}$$

so $\|u_i - v_i\|_{W_2^m} \leq C_4 B G_n^{-1}$ absorbing $|\mathcal{J}_i| \sqrt{4/3}$ into C_4 . By the additional regularity assumed in the premise, $\|\Phi_i^{(r)}\|_\infty \leq B_r < \infty$ for $r \leq m+1$, Φ_i is Lipschitz on $W_2^{m+1}(\mathcal{I}_\epsilon)$ with constant

$L_1 < \infty$ depending only on $B_1, \dots, B_{m+1}$ and the uniform bound B from **K.4(a)**:

$$\|\Phi_i(v_i) - \Phi_i(u_i)\|_{W_2^m} \leq L_1 \|v_i - u_i\|_{W_2^m} \leq C_5 B G_n^{-1} \quad (\text{C1.4})$$

For each $\psi_{i,l} \in W_2^{m+1}(\mathcal{I}_\epsilon)$, applying Theorem 12.8 (Schumaker, 2007) again:

$$\|\psi_{i,l} - \pi_n^{(l)} \psi_{i,l}\|_{W_2^{m+1}} \leq \sqrt{\frac{4}{3}} C_k B G_n^{-1} \quad (\text{C1.5})$$

Applying the chain rule (Adams and Fournier, 2003, Chapter 3) under **K.4(c)** across the decomposition as in Theorem 1:

$$\begin{aligned} \|\Phi_i - \pi_n \Phi_i\|_{W_2^{m+1}} &\leq \sum_{l=2}^{L_n} L^{L_n-l} \|\psi_{i,l} - \pi_n^{(l)} \psi_{i,l}\|_{W_2^{m+1}} \\ &\leq \sqrt{\frac{4}{3}} C_k B G_n^{-1} \left(\frac{L^{L_n-1} - 1}{L - 1} \right) \\ &\leq C_6 B G_n^{-1} \end{aligned} \quad (\text{C1.6})$$

where since $L_\psi \leq 1$ by **K.4(c)**, $\sum_{l=2}^{L_n} L_\psi^{L_n-l} \leq L_n - 1 = O(\log n)$ as established earlier for both $L_\psi < 1$ and $L_\psi = 1$ in the proof of **Theorem 1**, absorbed into the constant $C_6 = O(\log n) > 0$ as before. Combining bounds as before, we have:

$$\|\pi_n \theta_0 - \theta_0\|_{W_2^{m+1}(\mu)} \leq s_0 (C_5 + C_6) B G_n^{-1} + C'_{\bar{s}_0} O(n^{-1/2}) = C_{\bar{s}_0} G_n^{-1} + C'_{\bar{s}_0} O(n^{-1/2}) \quad (\text{C1.7})$$

where $\gamma > \frac{1}{4}$ delivers the $o(n^{-1/4})$ rate. $\square$

Proof of Theorem 2: In this proof we account for two components, the term arising from selection of (active) edges and the basis function approximation. The total parameter count is $p_n = r_n G_n$, where $r_n = L_n W_n^2 = O((\log n)^3)$ is the number of potential edges and G_n is the number of B-spline coefficients per edge.

For the former component, we have the term $2s_n \log \left(\frac{r_n}{s_n} \right)$ which is a generalization bound

of the combinatorial complexity of sparse function classes (by Lemma 17.5.1 of Cover and Thomas, 1991); highlighting the s_n active edges from the r_n potential edges in total. To see this, note that:

$$\log \binom{r_n}{s_n} \leq s_n \log \left(\frac{r_n}{s_n} \right) + s_n \leq 2s_n \log \left(\frac{r_n}{s_n} \right) \quad (\text{T2.1})$$

for $s_n < r_n/e$, which is ensured asymptotically by $s_n = O((\log n)^2)$ and $r_n = O((\log n)^3)$.

As for the latter component, the activation functions across active edges are approximated by G_n B-spline functions with $\ell_{1,2}$ -constrained coefficients: $\sum_{j=1}^{r_n} \|\mathbf{c}_j\|_2 \leq \Delta_n$ in $\mathcal{K}_n$. For the active edges, restrict to the s_n independent Euclidean balls $\mathcal{B}_j = \{\mathbf{c}_j \in \mathbb{R}^{G_n} : \|\mathbf{c}_j\|_2 \leq \Delta_n\}$. For each such ball, by Theorem 4.2.11 of Vershynin (2018), the ϵ -covering number satisfies:

$$\log \mathcal{N}(\epsilon, \mathcal{B}_j, \|\cdot\|_2) \leq G_n \log \left(\frac{3\Delta_n}{\epsilon} \right) \quad (\text{T2.2})$$

Summing over s_n active edges:

$$\log \mathcal{N} \left(\epsilon, \bigcup_j \mathcal{B}_j, \|\cdot\|_{L^2(\mu)} \right) \leq s_n G_n \log \left(\frac{3\Delta_n}{\epsilon} \right) \leq C_2 s_n G_n \log \left(\frac{\Delta_n}{\epsilon} \right) \quad (\text{T2.3})$$

where $C_2 > 0$ absorbs the $\log 3$ term. Combining the additive $o(1)$ term introduced by the (fixed unit) residual connections $h : \mathbb{R}^{W_n^{l-1}} \rightarrow \mathbb{R}^{W_n^l}$ with (T2.1) and (T2.3), we have the following bound:

$$\log \mathcal{N}(\epsilon, \mathcal{K}_n, \|\cdot\|_{L^2(\mu)}) \leq C_1 s_n \log \left(\frac{r_n}{s_n} \right) + C_2 s_n G_n \log \left(\frac{\Delta_n}{\epsilon} \right) + o(1), \quad (\text{T2.4})$$

where the dominant term is $C_2 s_n G_n \log(\Delta_n/\epsilon) = O((\log n)^2 \cdot n^\gamma \cdot \log(\Delta_n/\epsilon))$ under (2.8). $\square$

Proof of Corollary 2: Under **K.2(c)**, the envelope condition implies each $(\epsilon/2)$ -covering ball (under $\|\cdot\|_\infty$) generates an ϵ -bracket in $L^2(\mu)$ for any probability measure μ (van der Vaart and Wellner, 1996, Section 2.7). Therefore, $\log \mathcal{N}_{[]}(\epsilon, \mathcal{K}_n, \|\cdot\|_{L^2(\mu)}) \leq \log \mathcal{N}(\epsilon/2, \mathcal{K}_n, \|\cdot\|_{L^2(\mu)})$;

substituting $\epsilon/2$ into Theorem 2, we obtain the result with $\log(\Delta_n/(\epsilon/2)) = \log(2\Delta_n/\epsilon)$. $\square$

Proof of Theorem 3:

Since $\hat{\theta}_n \in \mathcal{K}_n$ by construction and $\mathcal{K}_n$ is compact under Δ_n , Theorem 1 of Chen and Shen (1998) applies directly (i.e., $\hat{\theta}_n = \arg \sup_{\theta \in \mathcal{K}_n} \mathcal{L}_n(\theta)$ is the exact maximizer under **K.2(c)**, such that Equation 2.1 of Chen and Shen, 1998 holds with $\tilde{\epsilon}_n = 0$). Under Assumptions **K.1–K.2**, their conditions A.1, A.2 and A.4 are satisfied (Condition A is outlined in its entirety in Section A.2). Condition A.1 is satisfied by all cases of **K.1**: trivially under **K.1(a)** ($\beta(j) = 0$ for $j > M$), under **K.1(b)** with block size $a_{n1} = \lfloor n^{1/2} \rfloor$, $a_{n2} = \lfloor n/(2a_{n1}) \rfloor$ (as required by Lemma 1 of Chen and Shen, 1998), and under **K.1(c)** by Corollary 1 of Chen and Shen (1998), which extends Theorem 1 to uniform mixing processes under the same block construction. A.2 is equivalent to **K.2(b)** and A.4 holds under **K.2(c)** with $\nu \in (0, 2)$, envelope $U(Z_t)$, and $E|U(Z_t)|^\kappa \leq C_2 < \infty$ for $\kappa \leq 2$, $\nu\kappa \leq 2$. For A.3, it is sufficient to verify the following integral condition:

$$\int_0^{\delta_n} \sqrt{\frac{\log \mathcal{N}_{[]}(\epsilon, \mathcal{K}_n, \|\cdot\|_{L^2(\mu)})}{n}} d\epsilon \leq C_3 \delta_n^2, \quad (\text{T3.1})$$

for $\delta_n = n^{-1/4-\eta}$ and some arbitrarily small $\eta > 0$. Using the dominant term from **Corollary 2**, this satisfies:

$$\begin{aligned} \sqrt{\frac{\log \mathcal{N}_{[]}(\epsilon, \mathcal{K}_n, \|\cdot\|)}{n}} &\leq \sqrt{\frac{C_2(\log n)^2 n^\gamma \log(2\Delta_n/\epsilon)}{n}} \\ &= \sqrt{\frac{C_2(\log n)^2 n^{\gamma-1} \log(2\Delta_n/\epsilon)}{1}} \end{aligned} \quad (\text{T3.2})$$

We decompose this by splitting the integral at $\epsilon_0 = n^{-1}$ and show that both terms are $o_P(n^{-1/4})$:

$$\int_0^{\delta_n} \sqrt{\frac{\log \mathcal{N}_{[]}(\epsilon, \mathcal{K}_n, \|\cdot\|_{L_2(\mu)})}{n}} d\epsilon = \underbrace{\int_0^{\epsilon_0} \cdots d\epsilon}_{(I)} + \underbrace{\int_{\epsilon_0}^{\delta_n} \cdots d\epsilon}_{(II)}$$

Term (I): We evaluate the integral via the substitution $u = \log(2\Delta_n/\epsilon)$, so $\epsilon = 2\Delta_n e^{-u}$ and $d\epsilon = 2\Delta_n e^{-u} du$. When $\epsilon = \epsilon_0 = n^{-1}$, $u = \log(2\Delta_n n) \leq C_\zeta \log n \equiv a$; as $\epsilon \rightarrow 0^+$, $u \rightarrow \infty$. Therefore:

$$(I) = \sqrt{C_2(\log n)^2 n^{\gamma-1}} \int_a^\infty u^{1/2} \cdot 2\Delta_n e^{-u} du \quad (\text{T3.3})$$

Integrating by parts with $v = u^{1/2}$, $dw = e^{-u} du$, we obtain:

$$\int_a^\infty u^{1/2} e^{-u} du \leq C a^{1/2} e^{-a} \quad (\text{T3.4})$$

for $a \geq 1$. Substituting $a = \log(2\Delta_n n)$ yields $e^{-a} = (2\Delta_n n)^{-1}$ and $\Delta_n = O(n^{\zeta_\delta})$:

$$\begin{aligned} (I) &\leq C(\log n) n^{(\gamma-1)/2} \cdot 2n^{\zeta_\delta} \cdot (C_\zeta \log n)^{1/2} \cdot \frac{1}{2n^{\zeta_\delta} \cdot n} \\ &= O(n^{(\gamma-1)/2-1} (\log n)^{3/2}), \end{aligned} \quad (\text{T3.5})$$

where ζ_δ cancels exactly. For $\gamma < \frac{1}{2}$: $\frac{\gamma-1}{2} - 1 < -\frac{3}{4} < -\frac{1}{2} - 2\eta$, therefore $(I) = o(\delta_n^2)$.

Term (II): for $\epsilon \in [\epsilon_0, \delta_n]$, bound $\log(2\Delta_n/\epsilon) \leq \log(2\Delta_n \cdot n) \leq C_\zeta \log n$ uniformly. Then:

$$\begin{aligned} (II) &\leq \int_{\epsilon_0}^{\delta_n} \sqrt{\frac{C_2(\log n)^2 n^\gamma \cdot C_\zeta \log n}{n}} d\epsilon = C(\log n)^{3/2} n^{(\gamma-1)/2} \int_{\epsilon_0}^{\delta_n} d\epsilon \\ &\leq C(\log n)^{3/2} n^{(\gamma-1)/2} \cdot \delta_n \end{aligned} \quad (\text{T3.6})$$

For (T3.1) to hold, we require $(II) \leq C_3 \delta_n^2$. This requires:

$$n^{(\gamma-1)/2+1/4+\eta} (\log n)^{3/2} \leq C_3, \quad (\text{T3.7})$$

where $\frac{\gamma-1}{2} + \frac{1}{4} + \eta = \frac{\gamma}{2} - \frac{1}{4} + \eta$. For (T3.7) to hold as $n \rightarrow \infty$, we require the left side to be bounded, i.e., $\frac{\gamma}{2} - \frac{1}{4} + \eta < 0$, which gives $\gamma < \frac{1}{2} - 2\eta$. Since $\eta > 0$ is arbitrarily small, the binding condition is $\gamma < \frac{1}{2}$, under which $n^{\gamma/2-1/4+\eta} (\log n)^{3/2} \rightarrow 0$ and (T3.7) is satisfied.

Since both Term (I) and Term (II) satisfy the fixed-point condition (T3.1) under $0 < \gamma < \frac{1}{2}$, Theorem 1 of Chen and Shen (1998) with block size $a_{n1} = \lfloor n^{1/2} \rfloor$, $a_{n2} = \lfloor n/(2a_{n1}) \rfloor$ (under **K.1(b)**, trivially under **K.1(a)**, and by Corollary 1 of Chen and Shen (1998) under **K.1(c)**) yields:

$$\|\hat{\theta}_n - \pi_n \theta_0\| = O_P(\delta_n) = o_P(n^{-1/4}) \quad (\text{T3.8})$$

since $\delta_n = n^{-1/4-\eta}$ for arbitrary $\eta > 0$. $\square$

A.2 Condition A and B (Chen and Shen, 1998)

Definition 3 (β -mixing). *A sequence $\{Z_t\}$ is β -mixing if its mixing coefficient $\beta(j) \rightarrow 0$ as $j \rightarrow \infty$, where given σ -fields $\mathcal{F}_{-\infty}^t, \mathcal{F}_{t+j}^\infty$ generated by $(Z_{-\infty}, \dots, Z_t)$ and $(Z_{t+j}, \dots, Z_\infty)$ (respectively):*

$$\beta(j) \equiv \sup_t \mathbb{E} \sup_{B \in \mathcal{F}_{t+j}^\infty} |P(B | \mathcal{F}_{-\infty}^t) - P(B)|$$

Definition 4 (ϕ -mixing). *A sequence $\{Z_t\}$ is uniform mixing (ϕ -mixing) if its mixing coefficient $\phi(j) \rightarrow 0$ as $j \rightarrow \infty$, where given σ -fields from **Definition 3** above:*

$$\phi(j) \equiv \sup_t \sup_{A \in \mathcal{F}_{-\infty}^t, P(A) > 0} \sup_{B \in \mathcal{F}_{t+j}^\infty} |P(B | A) - P(B)|.$$

A brief note on notation. Since the supremum over $\mathcal{K}_n$ is attained exactly under compactness and **K.2(c)**, the slackness condition from Chen and Shen (1998, Equation 2.1) used to define *approximate* sieve extremum estimators is equal to zero (written in terms of $\tilde{\epsilon}_n$ in this framework and discussed earlier⁵⁰ to avoid confusion; the KASN estimator is an *exact* sieve extremum estimator with $\tilde{\epsilon}_n = 0$). The reused term ϵ_n appearing in Conditions B.1-B.3 of Chen and Shen (1998) now corresponds to the sieve approximation error, bounded in **Theorem 1**.

⁵⁰See the proof of **Theorem 3** in the previous section.

Condition A:

A.1. $\{Y_i\}_{i=1}^n$ is a stationary β -mixing sequence with $\beta(j) \leq \beta_0 j^{-\xi}$ for some $\beta_0 > 0$, $\xi = \kappa - 2 > 2$.

A.2. For all small $\epsilon > 0$

$$\sup_{\{\theta \in \Theta_n : d(\theta_0, \theta) \leq \epsilon\}} \text{var}(l(\theta, Y_t) - l(\theta_0, Y_t)) \leq C_1 \epsilon^2.$$

A.3. Let $\mathcal{F}_n = \{l(\theta, Y_t) - l(\theta_0, Y_t) : d(\theta_0, \theta) \leq \delta, \theta \in \Theta_n\}$. There exists $\delta_n \in (0, 1)$ such that

$$\delta_n = \sup \left\{ \delta > 0 : \delta^{-2} \int_{b\delta^2}^{a\delta} H^{1/2}(w, \mathcal{F}_n) dw \leq C_2 n^{1/2} \right\}.$$

A.4. For any $\delta > 0$, there exist a constant $\nu \in (0, 2)$ and a measurable function $U_n(\cdot)$ such that

$$\sup_{\{\theta \in \Theta_n : d(\theta_0, \theta) \leq \delta\}} |l(\theta, Y_t) - l(\theta_0, Y_t)| \leq \delta^\nu U_n(Y_t),$$

with $\sup_n E[U_n(Y_t)]^\kappa \leq C_2$ for $\kappa > 2$.

Condition B:

B.1.

$$\sup_{\{\theta \in \Theta_n : \|\theta - \theta_0\| \leq \delta_n\}} \mu_n(r[\theta - \theta_0, Y] - r[P_n(\theta^*(\theta, \epsilon_n)) - \theta_0, Y]) = O_p(\epsilon_n^2).$$

B.2.

$$\sup_{\{\theta \in \Theta_n : 0 < \|\theta - \theta_0\| \leq \delta_n\}} \left(K(\theta_0, P_n \theta^*(\theta, \epsilon_n)) - K(\theta_0, \theta) - \frac{1}{2} [\|\theta^*(\theta, \epsilon_n) - \theta_0\|^2 - \|\theta - \theta_0\|^2] \right) = O(\epsilon_n^2).$$

B.3.

$$\sup_{\{\theta \in \Theta_n : 0 < \|\theta - \theta_0\| \leq \delta_n\}} \|\theta^*(\theta, \epsilon_n) - P_n(\theta^*(\theta, \epsilon_n))\| = O(\delta_n^{-1} \epsilon_n^2);$$

in addition,

$$\sup_{\{\theta \in \Theta_n : \|\theta - \theta_0\| \leq \delta_n\}} \mu_n \left(l'_{\theta_0} [\theta^*(\theta, \epsilon_n) - P_n(\theta^*(\theta, \epsilon_n)), Y] \right) = O_p(\epsilon_n^2).$$

B.4.

$$\sup_{\{\theta \in \Theta_n : \|\theta - \theta_0\| \leq \delta_n\}} \mu_n \left(l'_{\theta_0} [\theta - \theta_0, Y] \right) = O_p(\epsilon_n).$$

B.5.

$$n^{1/2} \mu_n \left(l'_{\theta_0} [U^*, Y] \right) \xrightarrow{P_{\theta_0}} \mathcal{N}(0, \sigma_v^2)$$

with

$$\sigma_v^2 = \lim_{n \rightarrow \infty} n^{-1} \text{var}_0 \left(\sum_{t=1}^n l'_{\theta_0} [U^*, Y_t] \right) > 0.$$

Proof of Theorem 4: By Theorem 2 of Chen and Shen (1998), it is sufficient to verify Conditions B.1-B.5 stated above. B.1 holds exactly under quadratic loss since the Taylor remainder is precisely $-\frac{1}{2}(\theta - \theta_0)^2$, which is $O(\epsilon_n^2)$ uniformly over $\|\theta - \theta_0\| \leq \delta_n$. B.2 holds under **K.2(a)** since $K(\theta_0, \theta) = \frac{1}{2}\|\theta - \theta_0\|^2$ exactly for quadratic loss, the condition reduces to verifying that $\|\pi_n \theta_0 - \theta_0\|^2 = O(\epsilon_n^2)$, which is the rate delivered by **Theorem 1**. Similarly, B.3 holds by **Theorem 1** for the first part and under **K.2(b)** by the Cauchy-Schwarz inequality for the score remainder. B.4 holds by applying Lemma 1 of Chen and Shen (1998) given the entropy bound of **Corollary 2** under **K.1**. B.5 holds by the CLT for β -mixing sequences (Yokoyama, 1980) under **K.1(b)**, trivially under **K.1(a)**, and by the CLT for ϕ -mixing sequences under **K.1(c)** (Doukhan, 1994), with long-run variance $\sigma_v^2 > 0$ guaranteed by non-triviality of v^* under **K.2(a)**. $\square$

A.3 KASN Optimization and Architecture Details

Define early stopping with patience ε_{tol} such that the computed estimator $\tilde{\theta}_n$ satisfies $\mathcal{L}_n(\tilde{\theta}_n) \geq \sup_{\theta \in \mathcal{K}_n} \mathcal{L}_n(\theta) - \varepsilon_{\text{tol}}$. By the triangle inequality and **K.2(a)**:

$$\|\tilde{\theta}_n - \theta_0\| \leq \|\tilde{\theta}_n - \hat{\theta}_n\| + \|\hat{\theta}_n - \theta_0\| = O(\varepsilon_{\text{tol}}^{1/2}) + o_P(n^{-1/4}).$$

Setting $\varepsilon_{\text{tol}} = o(n^{-1/2})$ ensures the early stopping error⁵¹ is $o(n^{-1/4})$ and does not affect the convergence rate of **Theorem 3**. This resulting condition is similar to the one in Chen and Shen (1998, Equation 2.1), however it does not concern the deterministic KASN class itself. In practice, the KASN estimator is optimized using the Adam optimizer (Kingma and Ba, 2014) with learning rate $\lambda_t \rightarrow 0$ satisfying $\sum_t \lambda_t = \infty$ and $\sum_t \lambda_t^2 < \infty$ (the standard stochastic approximation conditions of Robbins and Monro, 1951; see Zaheer et al. (2018) for the relevant discussion in the context of non-convex optimization problems). Under these conditions, Adam iterates converge to a stationary point of $\mathcal{L}_n$ over $\mathcal{K}_n$, and the early stopping criterion $\varepsilon_{\text{tol}} = o(n^{-1/2})$ ensures the resulting $\tilde{\theta}_n$ satisfies **Theorem 3** without affecting the convergence rate.

Numerical optimization is performed using PyTorch with GPU acceleration via CUDA when available (CPU otherwise). The local support properties of B-splines (see the definition of $\mathcal{I}_\epsilon$ in the main text) allow the per-edge spline evaluations to be expressed as a batched matrix operation. This vectorization ensures that increasing the grid resolution G_n does not require architectural changes to the computation graph, with runtime scaling linearly in G_n for fixed network width and depth.

The compositional structure of KAN architectures can be represented as follows, in con-

⁵¹See the discussion in the Section 2.3 about a similar condition that appears in Shen (1997) involving ϵ_n .

trast with the structure of a deep MLP:

$$\text{KAN}(\mathbf{x}) = (\Phi_{L_n-1} \circ \Phi_{L_n-2} \circ \cdots \circ \Phi_1 \circ \Phi_0) \mathbf{x}$$

$$\text{MLP}(\mathbf{x}) = (\mathbf{W}_{L-1} \circ \sigma \circ \mathbf{W}_{L-2} \circ \sigma \circ \cdots \circ \mathbf{W}_1 \circ \sigma \circ \mathbf{W}_0) \mathbf{x}$$

where σ denotes a fixed activation (e.g., ReLU) and Φ_l represents a collection of learnable univariate spline functions at layer l in the form of a matrix. In other words, the typical weight matrices on the edges of MLP structures are replaced with univariate functions in the form of B-splines.

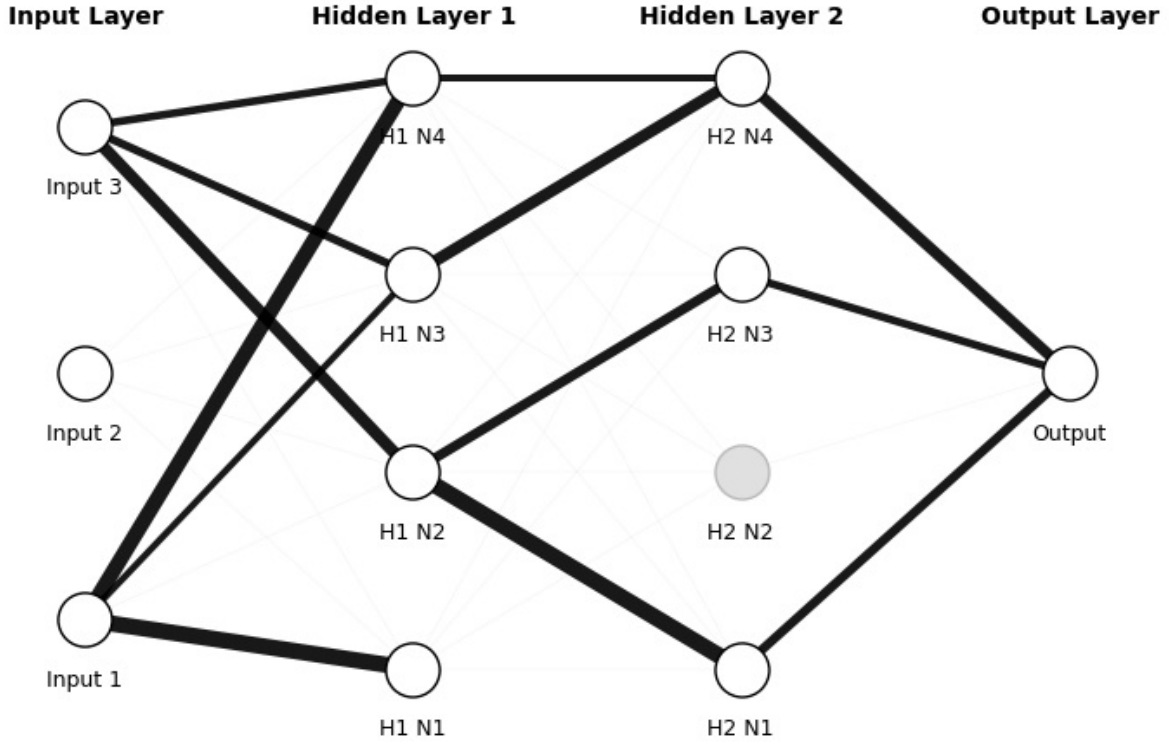


Figure 1: A KASN With Active Edges Highlighted

The difference between r_n and the number of *active* edges⁵² can be seen in Figure 1, where the solid lines represent learned activation functions with non-zero B-spline coefficients under ℓ_1 -regularization. In this example, the second input variable (x_2) does not contribute to the

⁵²Denoted by s_n and determined by s_0 in addition to the B-spline coefficient constraint Δ_n : another key parameter in our sieve space definition.

true target function in the DGP and there is an inactive node present in the second hidden layer, lacking any activation functions to sum as a result of the $\ell_{1,2}$ -regularization. Using r_n , we define our total parameters in the KAN as $p_n \equiv O(r_n G_n)$ as in Liu et al. (2024) where $p = O(LW^2G)$.

The fact that the network learns activation functions instead of weight matrices is what allows us to fix W_n across hidden layers in the first place, attempting to do so with MLP structures⁵³ is incompatible with our sieve estimation approach. Under a hypothetical framework incorporating MLPs with sieve estimation, ensuring that the MLP simultaneously possesses a bounded entropy and a degree of approximation power sufficient for high-dimensional estimation (with a dimension-independent convergence rate of $o_P(n^{-1/4})$ or faster) is an extremely challenging task. This is due to how the approximation power of an MLP is tied to its width and depth, which are typically required to grow with the sample size in order to achieve convergence rates comparable to the aforementioned minimax rates in nonparametric regression. As a result of this inherent dependency, it is difficult to control the metric entropy of such a sieve without inordinately restricting the approximation capabilities of the MLP.

In theory, the learnable activation functions $\phi(\cdot)$ can potentially include a basis function $b(x)$ in order to provide increased stability as part of our network initialization, as outlined in Liu et al. (2024). During training, the initial activation function would take on a weighted sum of the basis function and the B-spline function as follows:

$$\begin{aligned}\phi(x) &= w_b b(x) + w_s \text{spline}(x) \\ b(x) = \text{silu}(x) &= x/(1 + e^{-x}), \quad \text{spline}(x) = \sum_{g=1}^{G_n} c_g B_g(x)\end{aligned}$$

where $B_g(x) \equiv B_{g,k}(x)$ for a fixed spline order k and the weights (w_b, w_s) are updated during training in addition to the learned B-spline coefficients: with $w_b \rightarrow 0$ and $w_s \rightarrow 1$ early on

⁵³See Farrell et al. (2021) for a discussion on this class of “fixed width” networks, which are especially likely to struggle in high-dimensions.

in the training process, so that the resulting network derives its approximation capabilities purely from B-splines. Note that the basis function $b(x)$ is simply a combination of the ReLU and sigmoid activation functions, resulting in the “sigmoid linear unit”, or “SiLU” activation.

We find that across Monte Carlo simulations, this optimization approach can lead to a persistent degree of overfitting, one that cannot be addressed adequately by our novel regularization method once the transition to $w_s = 1$ is complete. In other words, the residual influence of $b(x)$ can persist in the learned edge function in a way that the $\ell_{1,2}$ penalty on spline coefficients $\{c_j\}$ does not directly penalize. To circumvent this issue, we instead introduce residual connections in the form of the identity mapping⁵⁴ $h : \mathbb{R}^{W_n^{l-1}} \rightarrow \mathbb{R}^{W_n^l}$, throughout the $l = 2, \dots, L_n$ hidden layers. The residual connection $h(x) = x$ passes the input from the previous hidden layer, improving stability during training by mitigating the vanishing gradient problem⁵⁵ in deep networks similar to the ReLU/SiLU activation functions. For this reason, the implementation of residual connections is common in many deep MLP architectures (inspired by He et al., 2016) and the growing literature on KANs (e.g., Yu et al., 2024). Furthermore, the presence of this identity mapping eliminates the necessity for carefully selecting appropriate spline initializations, allowing for a wide range of initialization techniques to be applied in practice. Given the relatively aggressive regularization approach of our framework, ensuring gradient stability is crucial with respect to learning the relevant true $\phi(x) = \sum_{k=1}^{G_n} c_k B_k(x)$ edge functions.

The B-spline functions $B_{j,k}(x)$ are defined recursively⁵⁶: for order k and a (non-decreasing) sequence of knot point t_j , the B-spline $B_{j,k}(x)$ is given by:

$$B_{j,k}(x) = \frac{x - t_j}{t_{j+k} - t_j} B_{j,k-1}(x) + \frac{t_{j+k+1} - x}{t_{j+k+1} - t_{j+1}} B_{j+1,k-1}(x)$$

⁵⁴As denoted in the definition of the KASN class $\mathcal{K}_n$ in equation (2.5).

⁵⁵Through bounding the gradient magnitude away from zero.

⁵⁶This definition is commonly referred to as the Cox-de Boor recursion formula: de Boor (1978).

with the base case for order $k = 1$ defined as:

$$B_{j,1}(x) = \begin{cases} 1 & \text{if } t_j \leq x < t_{j+1} \\ 0 & \text{otherwise} \end{cases}$$

In order to simplify the analysis in our key theorems/results, we fix the order of the B-splines as $k = 4$ (degree 3), with G_n intervals accordingly ($G_n + 1$ knot points in total: $t_0 < t_1 < \dots < t_{G_n}$). We note that in practice, KANs typically use cubic splines for a wide range of general applications, so this formalization does not incur additional costs or tradeoffs beyond in certain settings outside of the scope of nonparametric/semiparametric regression (e.g., symbolic regression). Furthermore, our theoretical results extend to any $k \geq m + 1$, although the sensitivity analysis of higher order B-splines (across a wide range of network parameters) evaluating their stability in this setting is left to future work.

In the KASN framework, we do not apply entropy regularization, a method that penalizes the average magnitude of the activation functions commonly applied in KANs (see the approach outlined in Liu et al., 2024, Appendix C therein). While this may assist in reducing the number of redundant functions in general KAN architectures (in settings where $n \gg d$), this form of regularization effectively distributes the task of learning the structure of f_0 across the many potential edges. This ultimately results in dense networks that will fail to prevent the explosion of the metric entropy in high-dimensions, unlike under the structural sparsity introduced in this framework. Instead, our constraint Δ_n promotes the opposite (in addition to penalizing entire vectors of B-spline coefficients across edges under our regularization method), forcing the network to use each active edge s_n as efficiently as possible *without* sacrificing approximation power.

Under our proposed regularization method (that is novel for KAN architectures) and pruning based on the Δ_n/r_n threshold for active edges, KASNs can be viewed as a deep compositional generalization of the Multivariate Adaptive Regression Splines (MARS) estimator (Friedman, 1991). Unlike MARS, which requires manual specification of the max-

imum degrees of interaction⁵⁷, KASNs hierarchically learn complex interactions through a deep composition of univariate B-splines. The KART guarantees this representation exists, with Assumption **K.3(a)** imposing the sparse structure that controls combinatorial complexity at an $O(s_n) = O((\log n)^2)$ rate. Furthermore, the Δ_n/r_n threshold used to prune univariate edges is a sieve analogue of the backward deletion procedure (guided by Generalized CV) introduced by Friedman (1991), one that is formally derived from the entropy bound (**Theorem 2**) rather than heuristically.

A.4 Conditional Density Estimation Example

In an earlier study addressing the CoD and potential avenues for circumventing it in the context of ReLU DNNs⁵⁸, Mhaskar and Poggio (2016) define *relative dimensions*. This concept provides quantitative measurements of sparsity tied to parameters of the DNN and leverages compositional function assumptions to provide a plausible explanation for the improved performance of DNNs relative to “shallow” networks (single hidden layer feedforward networks, or SLFNs). We contrast this with our proposed KASN using a conditional density estimation example in this section.

Suppose we seek to employ a deep MLP in a sparse estimation problem involving covariates with high-dimensions and naively attempt to implement ℓ_1 regularization (or the more common “elastic net” method combining it with ℓ_2 regularization). This applies a penalty $\sum_{i,j} |w_{ij}|$ on all weights across every layer, in theory reducing the contributions of weights significantly influenced by irrelevant features. Achieving this in practice with single hidden layer networks is more feasible relative to DNNs⁵⁹, due to the complex interconnected nature of the weights in the latter architecture. Consider the following function:

$$f_0(x_1, x_2, \dots, x_{10}) = -3.0 + 2 \sin(x_1) + 0.1(x_2^2) + \epsilon, \quad \epsilon \sim N(0, 0.25)$$

⁵⁷In order to control the complexity of basis functions.

⁵⁸With a focus on deep convolutional networks, see LeCun et al. (2015) for an overview.

⁵⁹See Shen et al. (2023).

where the true density of the relevant variables (x_1 and x_2) takes on a bullseye shape, $d = 10$ and $n = 10,000$.

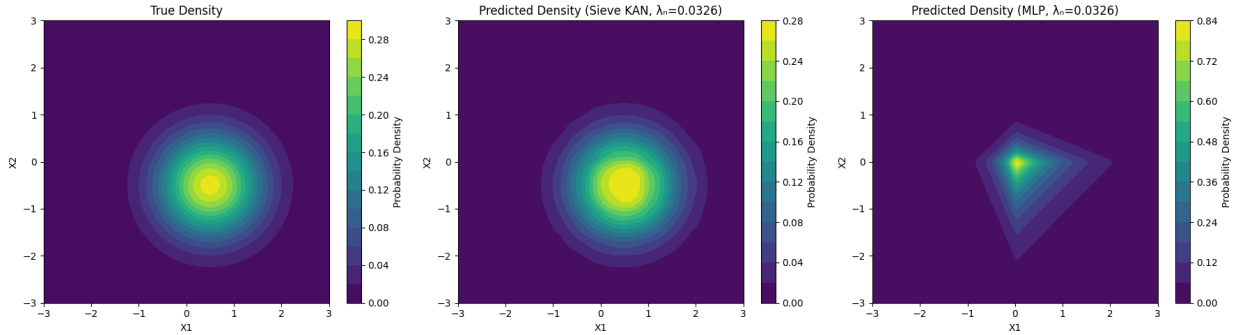


Figure 2: CDE Example Predicted Densities

As seen in Figure 2, a deep MLP (in this case, a ReLU DNN) applying ℓ_1 regularization will struggle to learn the shape of the true density. However in the absence of this regularization, the MLP is more prone to overfitting in noisy DGPs⁶⁰, highlighting a key tension in the potential application of deep MLPs in high-dimensions. As for our KASN, the predicted density is much closer to the true density in terms of shape and scale, despite relying only the learned univariate functions preserved under ℓ_1 -regularization. We note that due to how $n \gg d$, this prediction can easily be improved further by increasing λ_n and/or G_n through γ (initially set to $\gamma = 0.3$ to obtain a grid size similar to those in DGP #1 and #2 in the earlier simulations). In line with the philosophy of truly data-adaptive approaches to nonparametric estimation, these parameters can also be automatically optimized using cross-validation⁶¹ in a manner similar to the Lasso penalty selection offered by packages across various programming languages.

A.5 Shallow Sieve Networks (Chen and White, 1999)

Let d_x denote the dimension of the covariates. Let $\mathcal{B}_{d_x}^m$ denote the weighted Sobolev space comprising all functions on $\mathbb{R}^{d_x}$ whose partial derivatives up to order m are both continuous

⁶⁰See Schmidt-Hieber (2020)

⁶¹See the documentation for several examples of this in practice, in addition to standard hyperparameter tuning; e.g., setting the learning rate.

and uniformly bounded (see the definitions in Section 2). The target function $f : \mathbb{R}^{d_x} \rightarrow \mathbb{R}$ is assumed to admit a Fourier representation such that it belongs to the functional class:

$$\mathcal{F}_{d_x}^{m+1} = \left\{ f : \mathbb{R}^d \rightarrow \mathbb{R} \left| f(x) = \int \exp(ia^\top x) d\sigma_f(a), \|\sigma_f\|_{m+1} \equiv \int I(a)^{m+1} d|\sigma_f|_{\text{tv}}(a) < \infty \right. \right\} \quad (\text{A.1})$$

This assumption imposes a general smoothness constraint (ensuring the Fourier transform possesses a bounded first moment) required by the class of SLFNs utilized in Chen and White (1999), denoted by $\mathcal{G}_n \equiv G_{d_x}^m(\psi, r_n, \Delta_n)$ and defined as:

$$\mathcal{G}_n = \left\{ g : g(x) = \sum_{j=1}^{r_n} v_j l(a_j)^{-m} \psi(a'_j x + w_j), \right. \quad (\text{A.2})$$

$$\left. a_j \in \mathbb{R}^{d_x}, (w_j, v_j) \in \mathbb{R}, \sum_{j=1}^{r_n} |v_j| \leq \Delta_n \right\}$$

Here, $\psi \in \mathcal{B}_1^m$ represents the selected activation function (satisfying a Hölder condition) that is k -finite for some $k \geq m$. The parameter vector $\omega_j \equiv (a_j, w_j, v_j)$ contains the network weights, where r_n specifies the number of hidden units as discussed earlier in this paper. For any weight vector a_j , the term $l(a_j) \equiv \max\{(a'_j a_j)^{\frac{1}{2}}, 1\}$ normalizes the input layer weights, thereby controlling model complexity and enforcing the required smoothness of the approximation.